Uganda National Roads Authority

ROADS AND BRIDGES

SITUATIONAL ANALYSIS REPORT

Uganda National Roads Authority

Network Planning Department

August 2022

PLOT 3-5, NEW PORTBELL RD, NAKAWA UAP BUSINESS PARK.

P.O BOX 28487 KAMPALA, UGANDA

Contents

Executive Summary

The Directorate of Network Planning and Engineering is mandated to; Collect and update road network and bridge asset data, Maintenance of Road, and Bridge data bank and Road Management Systems; Formulate short and long-term bridge and road investment plans; Collect road condition and traffic data to support development and maintenance programming of the network among other duties. The Network planning department executes these particular tasks.

In that regard, annual data collection studies comprising of inventory and condition assessment were undertaken. The principal objective of this study was to prepare a situational assessment report that accounts for the physical condition and usage of the road network for FY 2021/22.

A total of 5,675km of paved roads and 12,629km of unpaved roads were assessed. The average condition of the paved roads was classified as “good” and unpaved roads based on the visual condition ratings as “fair”. The percentage of unpaved roads in the poor and very poor category has dropped to 11% in 2022 as compared to 18% in the previous assessment of 2021.

There is a total of 816 structures with 423 bridges and 393 major culverts on the national network. The average overall condition of structures is rated “good” out of which 232 bridges and 306 major culverts account for the good and satisfactory condition.

Of the 298 No traffic count stations located on the network, a total of 191No traffic count stations were surveyed. 61No and 130No stations on the paved and unpaved network respectively. The paved road network carries an ADT of 5,009 while the unpaved road network carries 1,391. Motorcycles or scooters (boda-bodas) are 59% of the total traffic by volume while Medium buses and large buses are the least dominant vehicle class composing of 1% of the traffic.

1. Introduction

This document provides the results of the situational analysis which is the first part of the National Roads Investment plan. It provides an understanding of the composition of the National road network, traffic information as well as its current condition of the Asset. A precise understanding of the current condition of a network is the first major step in infrastructure planning. Therefore, the findings herein will be inputs in the needs analysis whose outputs will be; preventive maintenance, rehabilitation and upgrading needs of the UNRA road network, also referred to as an Investment Plan or National Road Investment Plan (NRIP).

The purpose of this analysis was to report on the current status of the asset and further prepare inputs into the needs analysis for a multi-year investment plan for the road network. The analysis for this report was done using programmable spreadsheets and dTIMS v9.5 software customised with the Highway Development and Management (HDM-4) pavement performance prediction algorithms.

2. The UNRA Road Network

- *The National Network is 21,120 km in total.*
- *27% (5,745km) is paved, and 73% (15,375km) unpaved.*
- *5,675km (99%) of paved roads were analysed in this report.*
- *12,629km (82%) of unpaved roads were analysed in this report.*
- *There are 816No Bridges and structures on the Network.*

Network Classification and Length

The total length of the national road network is 21,120 km. Of this 5,745 km is paved and 15,375 km remains unpaved. Furthermore, according to functional classification, 12.3% of this network is Class A, 13.5% Class B, 73.7% is Class C and 0.6% is Class M as indicated below.

Table 2-1: Classification of the UNRA Road Network

Functional Classification	Paved	Unpaved	Length (Km)	% Total	Description
A	2601	0	2601	12.3%	International Trunk Roads
B	1493	1349	2843	13.5%	National Trunk Roads
C	1533	14026	15559	73.7%	Primary Roads
M	117	0	117	0.7%	Expressways
Total	5745	15375	21120	100%	

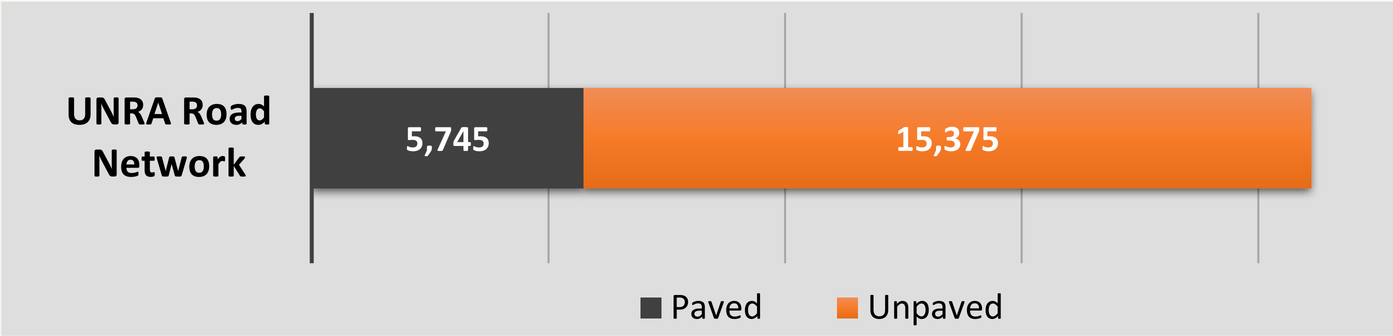


Figure 2-1: Split between Paved (27%) and Unpaved (73%) Road lengths of UNRA Network (2022)

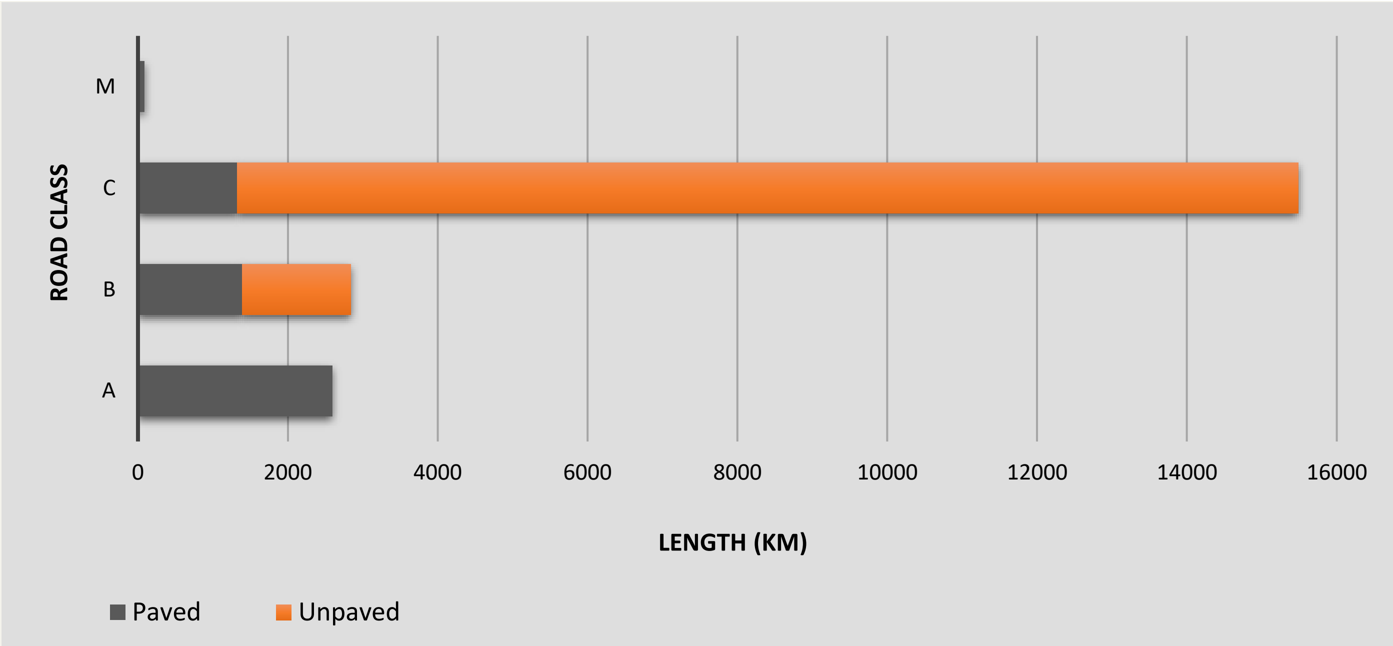


Figure 2-2: Functional Classification of UNRA Road Network

The national network has increased by 100km from 21,020km to 21,120km in total lane length. This stems from the addition of 8No new links, completion of new dual sections and increment in length of some roads. Tables 3.2 and 3.3 below summarize the network changes.

Table 2-2: Changes in length from 2020/21 to 2021/22

Link Name	New Length (km)	Difference in Length (km)	Comment
Entebbe Airport (Dual)	12.9507	7.5177	Dual section from Statehouse to Mpala interchange (7.5177km) added
Kasana - Bugobero	7.63	7.63	New on network
Access to Kigungu	10	5.956	Increment in length by 5.956km
Nkondo - Nalweyo	30.21	30.21	New on network
Zengebe-Ferry-Access	3.025	3.025	New on network
Kikubampanga - Lutabayima - Seganga	5.801	5.801	New on network
Access to Tumu hospital	10.195	10.195	New on network
Teokur - Aboo	13.21	13.21	New on network
Rhino Camp Ferry Access	5.85	5.85	New on network
Old -access-to-Isimba-dam	3.96	3.96	New on network
Northern Bypass	42.046	17.563	Changed to M20 with a total length of 42.026km. New dual-lane of Namboole-Gayaza (12.351km), Hoima road-Busega (5.233km) added
Kampala - Entebbe Expressway	75.136	-10.664	Changed length to 75.13622km. Excluded the section from Mpala interchange - Entebbe airport of single lane length 12.9507km
Total change in length		**100.254**	

Table 2-3: Changes in Length (2008 – 2022)

Category/Year	2008/9	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2019/20	2020/21	2021/22
Paved (Km)	2916	3119	3264	3317	3490	3795	3919	4157	4257	4636	5015	5398	5528	5745
Unpaved (Km)	7137	7311	17120	17683	17510	16749	16625	16387	16287	15908	15841	15612	15492	15375
Total	10053	10430	20384	21000	21000	20544	20544	20544	20544	20544	20856	21010	21020	21120

Length Investigated

The total length of the national road network investigated in this analysis is based on the length of roads with complete assessment data in 2021/22. 5,675km of paved roads have updated measurements for roughness, Rutting, video, Geometry and GPS data while 5,449km of paved roads and 12,629km of unpaved roads have complete visual assessment data. In total 18,304km (87%) of the total length of roads were investigated which represents 99% of paved roads and 82% of unpaved roads.

Note that although it is preferable that data for the full road network be available for any analysis, this was not possible due to; the insecurity in the Karamoja region that made assessments in Kotido and Moroto stations impossible; delay in contract signing of Lot 1 (central region) due to unavailability of funds; roads under construction at the time of surveys; and changes or corrections to the roads register.

The graph below details the length investigated for FY 2021/22.

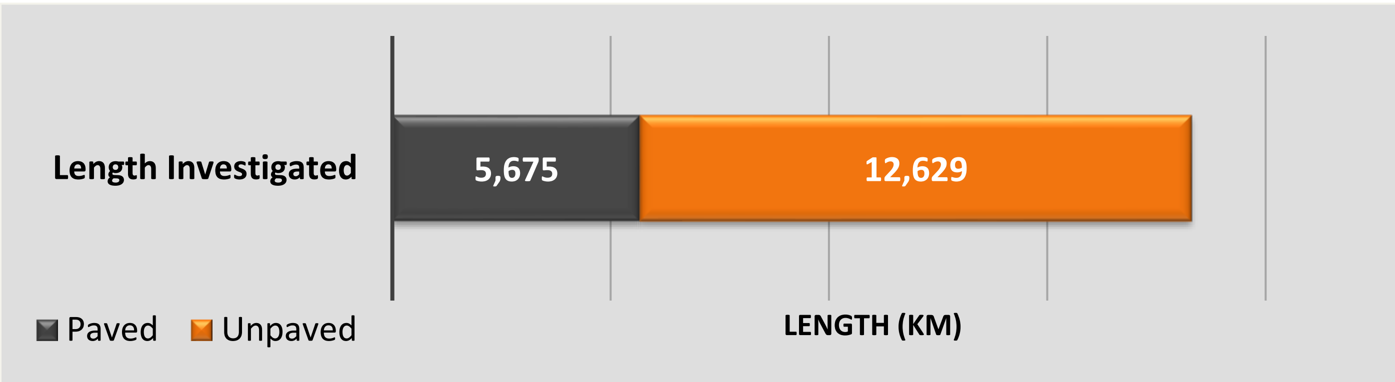


Figure 2-3: Paved and Unpaved Road lengths investigated (km), 2022

Bridges and Structures

Based on the assessment of FY 2021/22, the total number of structures on the network is 816. These are categorized as Bridges and Major Culverts as the two have different maintenance needs and life spans. For this cycle (FY2021/22) inventory and condition assessment of these structures was done for the Northern, Southern and Western regions. Historic data was used for the unassessed structures. Validation of this data has been ongoing and the numbers are likely to change due to the construction of new structures and upgrading of roads. Inventory of some of the roads under upgrade has not yet been covered.

The graphs below detail the number of structures on the network.

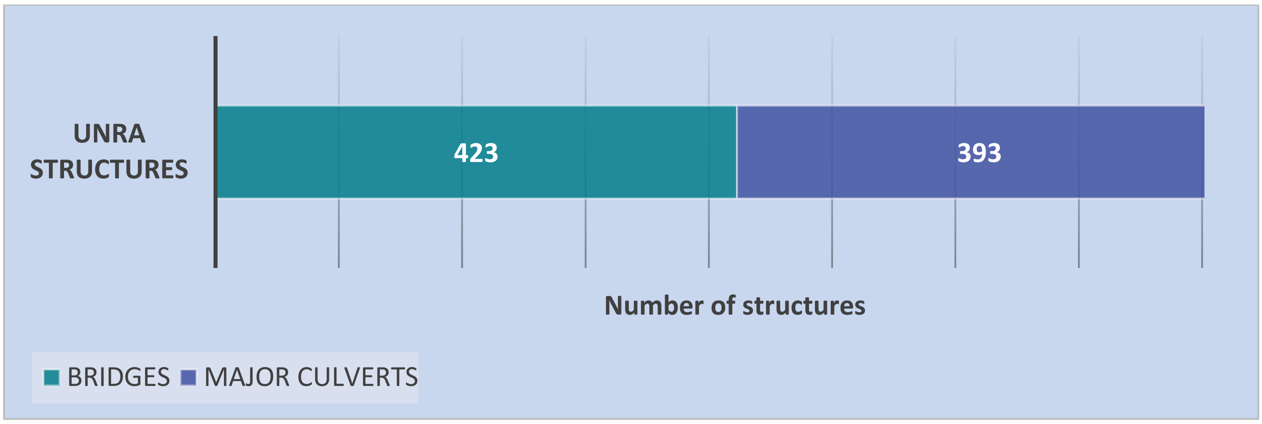


Figure 2-4: Split between Bridges and Major Culverts on the UNRA Network, 2022

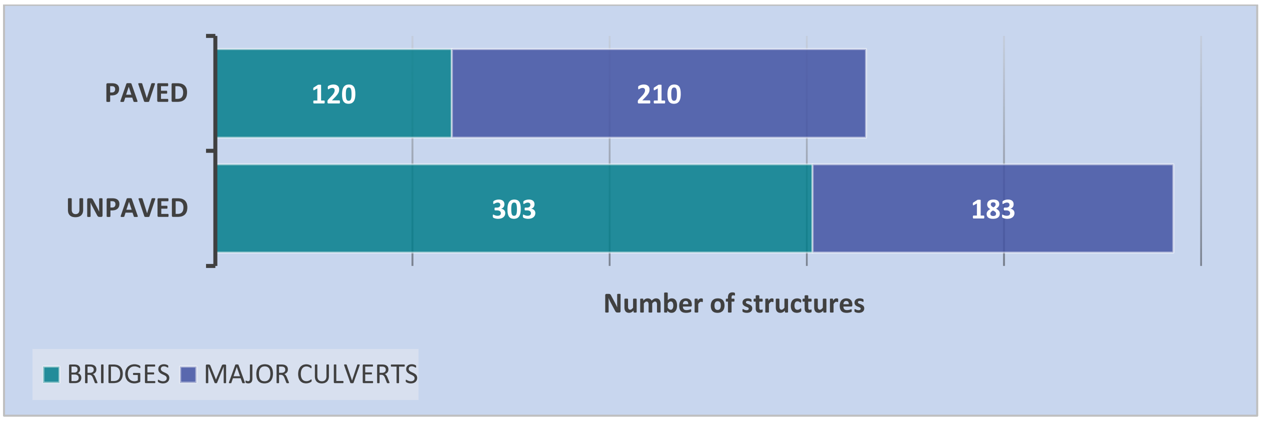


Figure 2-5: Split between Bridges and Major Culverts on the Paved and Unpaved Network, 2022

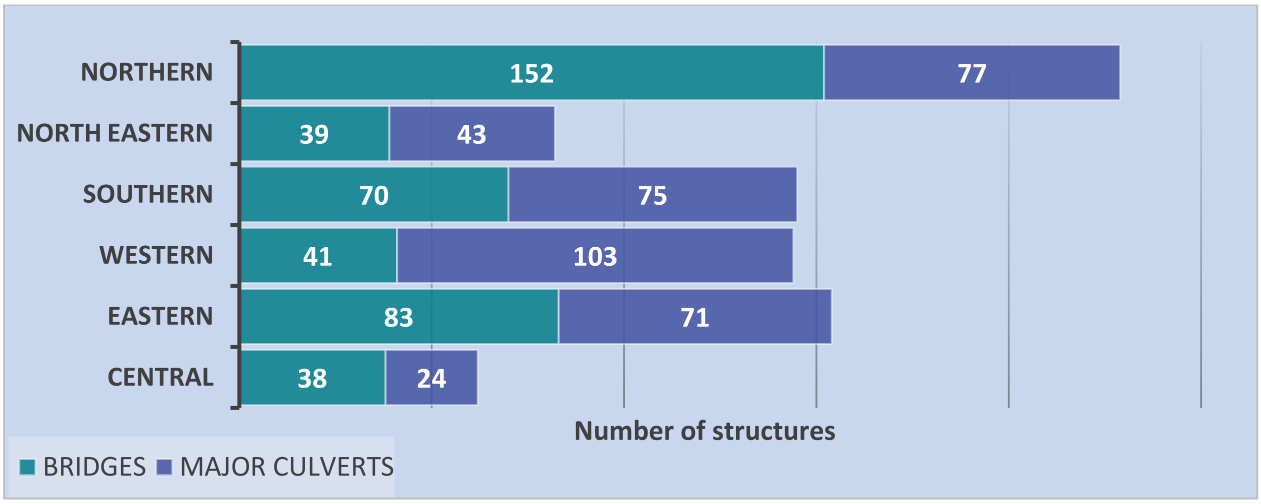


Figure 2-6: Split between Bridges and Major Culverts per region, 2022

From Figure 2-5 above, it can also be observed that the Northern region followed by the Eastern regions have the highest number of structures while the Central region has the smallest number of structures. This has an impact on the funding levels for routine maintenance of structures per region.

3. Traffic Information

The Average Daily Traffic (ADT) on the surveyed road network is 2,547 vehicles per day inclusive of motorcycles and 1,039 vehicles per day exclusive of motorcycles.
The Average Daily Traffic (ADT) on the paved road network is 5,009 vehicles per day inclusive of motorcycles and 2,404 vehicles per day exclusive of motorcycles.
The Average Daily Traffic (ADT) on the unpaved road network is 1,391 vehicles per day inclusive of motorcycles and 398 vehicles per day exclusive of motorcycles.

Traffic data was collected by both internal staff and Data Collection Consultants in the third and fourth quarter of FY2021/22 at 191No out of the 298No selected traffic counting stations on the network. The central region with the largest number of count stations (102No.) was only partially surveyed with 20 count stations surveyed by the In-house team. This was due to a delay in the contract signing of Lot 1. The count stations in Kotido and Moroto maintenance stations were also not surveyed due to the increased insurgence in the Karamoja region. The data collected was analysed and expressed as the Average Daily Traffic (ADT), travelling on each road link surveyed which represents the vehicle traffic usage.

32% (61No.) of the manual counting stations investigated are located on paved roads and 68% (130No.) on unpaved roads. From the analysis, the ADT on paved roads is 5,009 vehicles per day inclusive of motorcycles whereas that on unpaved roads is 1,391 vehicles per day inclusive of motorcycles. This indicates that paved roads carry about threefold as much traffic as unpaved roads (see Figure 3-1). A focus on the maintenance and rehabilitation of paved roads will therefore influence more road users per kilometre than unpaved roads.

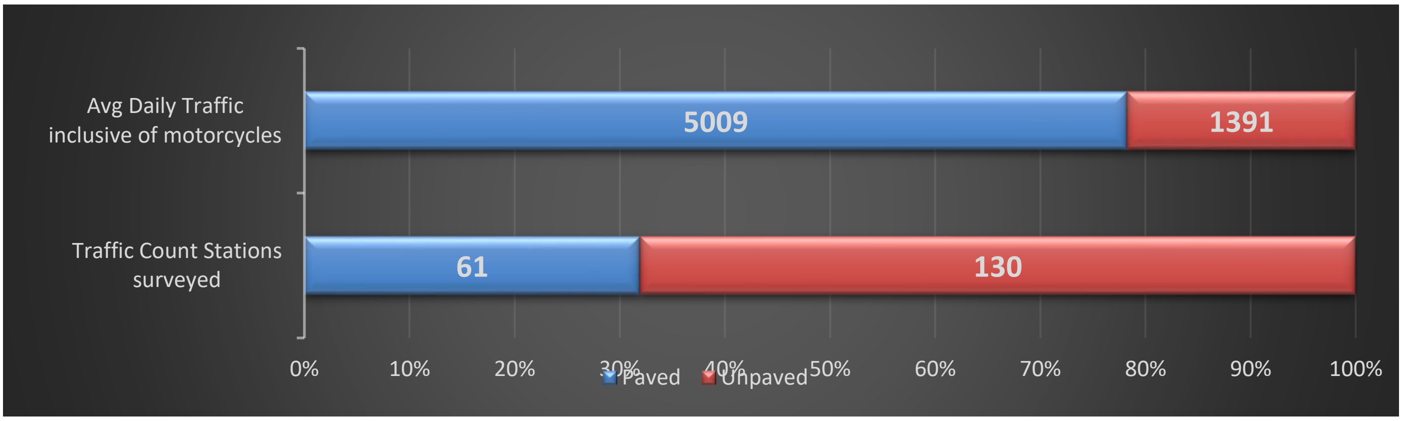


Figure 3-1: Traffic Count Stations investigated Vs Average Daily Traffic, 2022

3.1 Traffic on the network

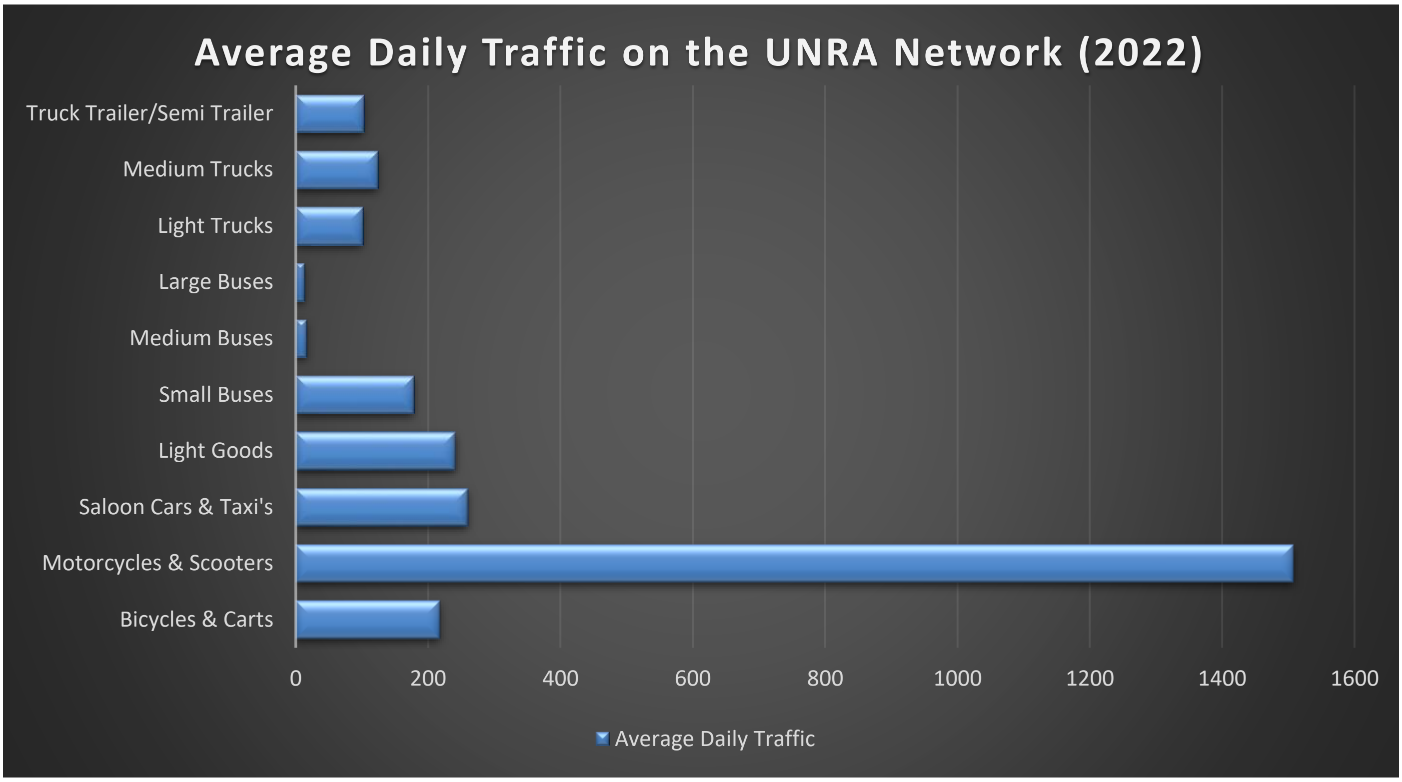


Figure 3-2: Average Daily Traffic inclusive of motorcycles per Vehicle Type, 2022

From the figure above, it is observed that motorcycles & scooters are by far the most dominant means of transport on the road network with an ADT of 1,508 followed by Saloon cars & taxis (ADT 260), Light goods (ADT 242), Bicycles & Carts (ADT 218). Large buses (ADT 13) and Medium buses (ADT 16) have the smallest volumes on the network.

Figure 3-3 below shows the average motorised traffic per vehicle type on the road network

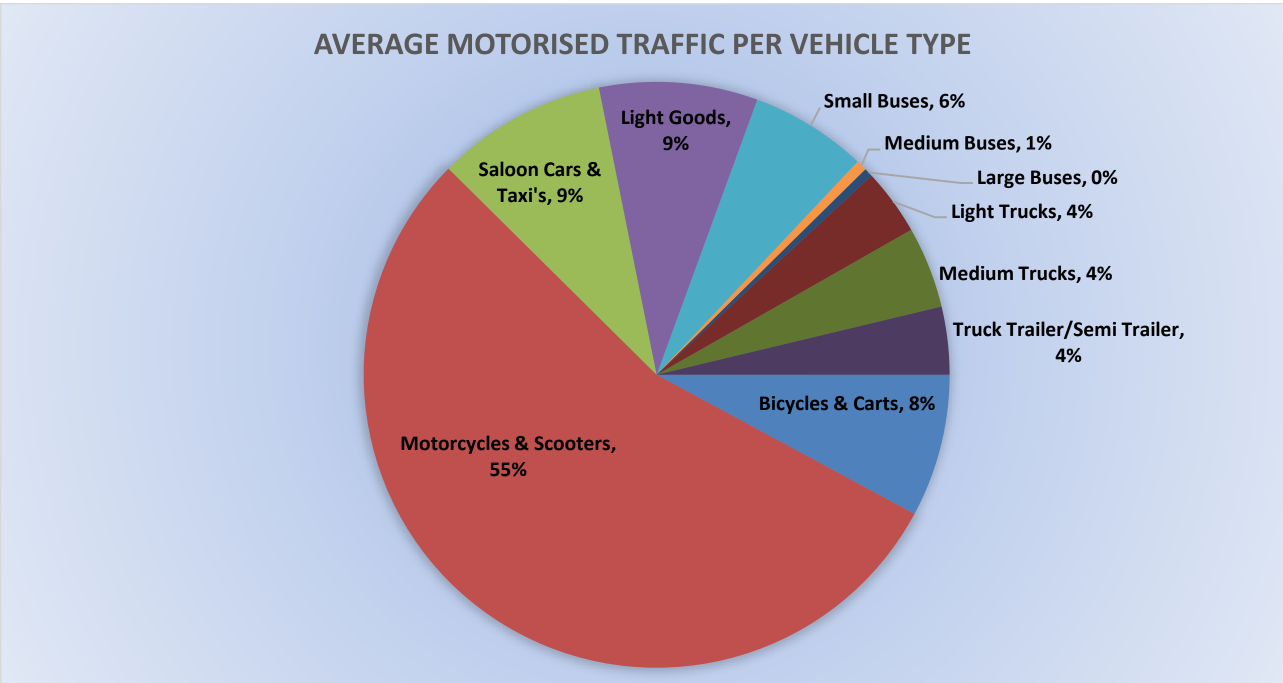


Figure 3-3: Average motorised traffic per vehicle type, 2022

3.2 Traffic per road surface type

The distribution of traffic volumes for paved and unpaved roads is given in Figure 3-4 and 3-5 respectively.

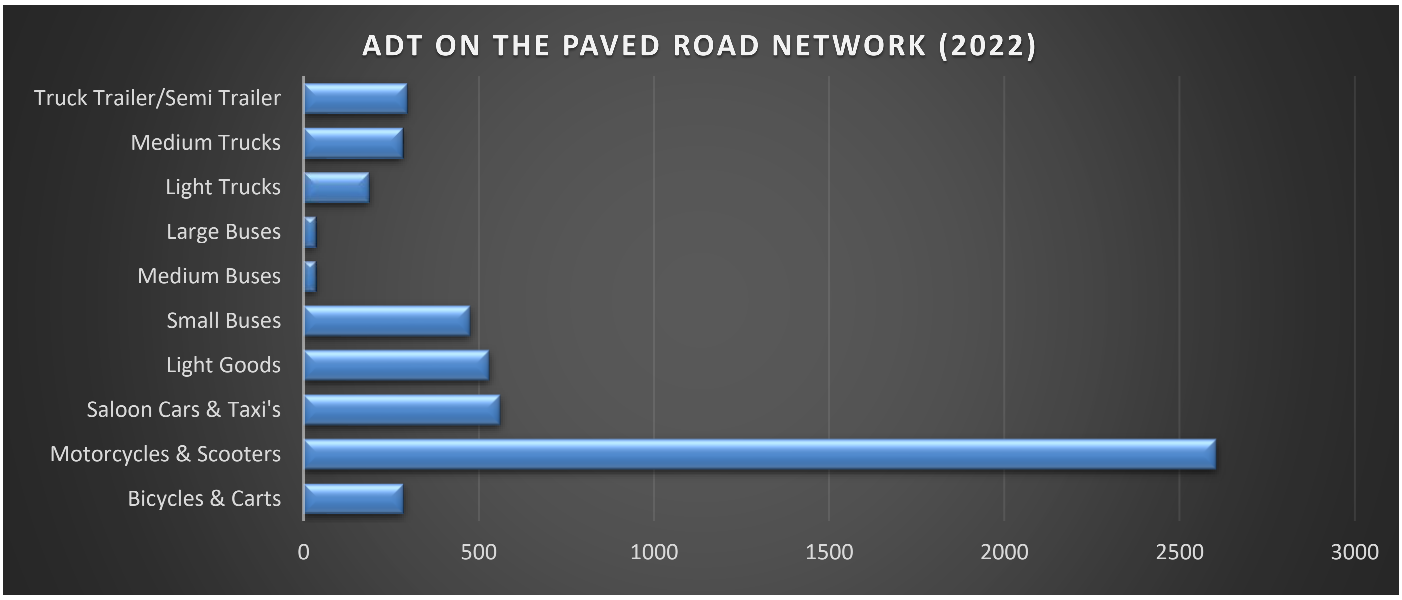


Figure 3-4: Average Daily Traffic on paved roads, 2022

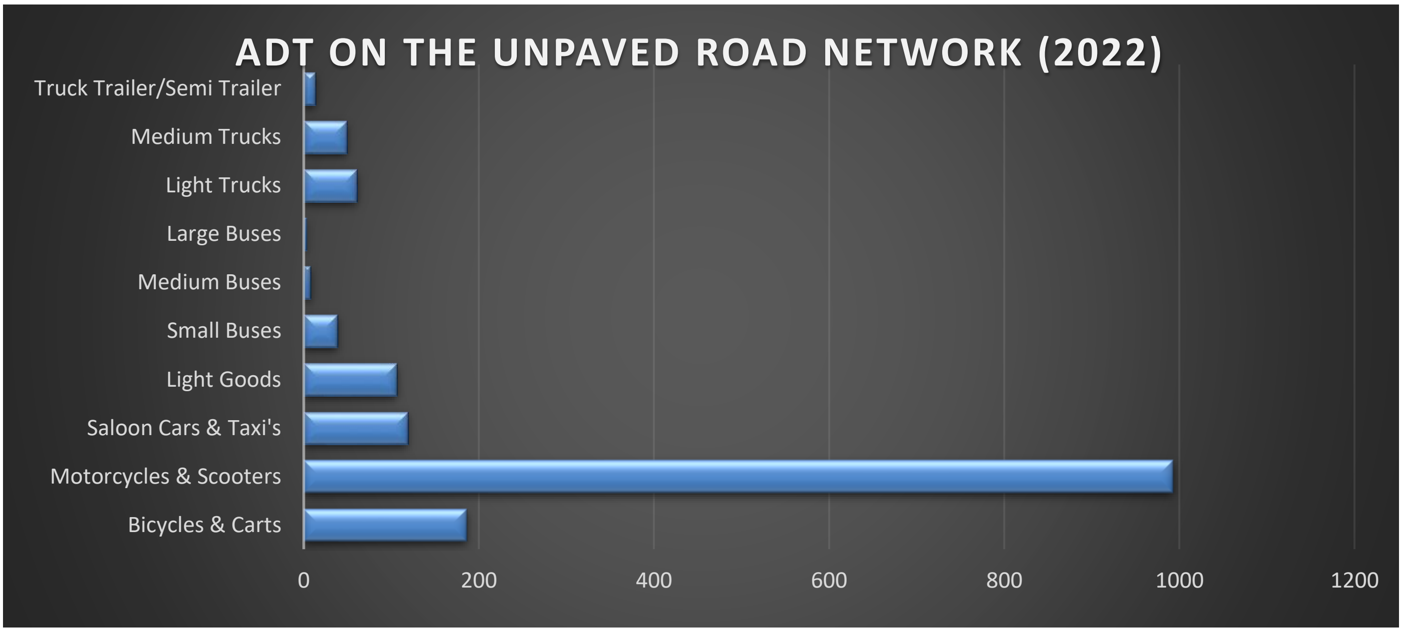


Figure 3-5: Average Daily Traffic on unpaved roads, 2022

The figures above indicate that Motorcycles & Scooters also have the highest volumes on the road network with an ADT of 2,606 and 993 on the paved and unpaved roads respectively. Medium buses and large buses have the smallest volumes with a total ADT of 69 and 11 on the paved roads and unpaved roads respectively.

4. Paved Roads Network Condition

- *The average paved road VCI based is 79%.*
- *Moderate to severe 'All Cracking' is existent on 24% (1,317km) of the paved roads.*
- *Moderate to severe 'Wide Cracking' is existent on 22% (1,177km) of the paved.*
- *16% (863km) of roads have severe to moderate ravelling (surface disintegration) and immediate maintenance is required on these roads.*
- *There is severe to warning roughness on 7% (410km) of the paved roads.*
- *6% (342km) of paved roads are in a moderate to severe condition due to the occurrence of potholes. Isolated potholes occur on a further 16% (892km) of the network.*
- *24% (1,316km) of the road network exhibits rutting in a moderate to severe state.*
- *Overall, the paved road network is in good to fair condition with less than 10% in poor and very poor condition.*

4.1 Background to the VCI

The assessment data, expressing the carriageway condition through the assessment ratings of pavement defects are utilised to calculate a single Visual Condition Index (VCI). This VCI is calculated for each paved road segment (typically of 1 km length) that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents very good condition. During this analysis the VCI is used as follows:

- An average network VCI is calculated to represent the average condition of the paved road network or any sub-network (e.g., regional network). This average network VCI is weighted by length and is used to describe/investigate the average condition of the road network/sub-network.
- The VCI is furthermore used to categorise the condition of a paved road segment, the paved road network or sub-network into one of five condition categories. The condition categories adopted are:

  - Very Good = 86% to 100%
  - Good = 71% to 85%
  - Fair = 51% to 70%
  - Poor = 31% to 50%
  - Very Poor = 0% to 30%

4.2 Visual Assessment and Distress Graphs for Paved Road network

The Surface condition of paved roads was assessed by both visual inspection and by pavement surveillance measurement equipment. UNRA staff executed paved visual assessments and paved roughness measurements in 2021/22. The Pavement Management System (PMS) of the UNRA RMS was updated with this data after data validation.

Figure 5-1 is a schematic representation of the distress items showing the total percentage of paved roads rated as 1 (very good or non-existent), 2 (slight, good or isolated occurrence), 3 (discernible, fair or moderate occurrence), 4 (warning, poor or general occurrence) and 5 (severe, very poor or extensive occurrence). These ratings can indirectly be related to the proportion of the paved road network with none, isolated, moderate, warning and severe problems.

According to international standards, a "healthy" road network has less than 10% poor and very poor roads. Attention should thus be given to the distresses and condition ratings exceeding this benchmark for warning and severe problems. According to the visual assessments and automated measurements the following conclusions can be made with regard to the paved roads:

Cracking

- Moderate to severe 'All Cracking' is noted on 24% (1,317km) of the paved roads with 28% (1,510km) exhibiting isolated cracking and 48% (2,624km) exhibiting no cracking.
- 22% (1,177km) of the paved network presents moderate to severe wide cracking, another 22% (1,225km) presents isolated wide cracking and 56% (3,049km) presents no wide cracking. Wide cracking is a particular concern because these cracks indicate the disintegration of the structural layers in the wheel paths. W ide cracks also initiate the formation of rutting. Preventive measures should be considered as soon as possible to delay further deterioration with the development of secondary distresses.

Ravelling

- Ravelling is the loss of aggregate material from the road surfacing. In seals, this manifests through stone loss, in asphalts through the disintegration of the surfacing.
- 6% (318km) of roads have severe to warning ravelling and immediate maintenance is required on these roads.
- 10% (545km) display moderate ravelling. Future maintenance of roads with moderate ravelling is crucial since it accounts for the percentage of roads which could easily deteriorate to warning condition.

Roughness (measured)

- ROMDAS surveys for roughness (IRI) and rutting were done on 5,675km (99%) of the paved road network.
- From the IRI surveys, 7% (410km) of the road network has severe to warning roughness. These roads typically require light or heavy rehabilitation. 28%

(1,590km) have moderate roughness and 65% (3674km) have very good and good roughness. Although some of these roads might not require immediate attention it is crucial to monitor the condition continuously and apply the right treatment at the right time to delay further deterioration.

Rutting	▪ 25% (1,347km) of the road network exhibits rutting in moderate to severe state, 59% (3,234km) with isolated rutting and 16% (870km) with no rutting.
Bleeding	▪ 4% (237km) of the paved network exhibits bleeding in severe to warning states with 24% (1,313km) having isolated bleeding and 72% (3,901km) having no bleeding.
Width Loss	▪ There is very minimal width loss (due to edge break) of the pavement on the network. 99% (5,420km) of the network exhibits no edge break.
Edge Drops	▪ There is also very minimal edge drop on the network with 99% (5,405km) of the network with no edge drop.
Shoulder condition	▪ The 2m width is available to the road user as an emergency stopping area. 66% (3,579km) have no defects, 17% (952km) have isolated problems and 17% (919km) are in moderate to severe condition.
Drainage on road	▪ This refers to surface drainage and the ability of road to drain water from the surfacing. Generally, not a problem with 84% (4,572km) in good condition, only 13% (698km) in moderate condition and 3% (181km) in severe condition.
Drainage off road	▪ Referring to the condition of side drainage, mitre drains and cross drainage underneath the road. 17% (899km) of the paved road network in warning to severe conditions, 50% (2,703km) in a warning state, 30% (1,616km) with isolated drainage problems and 4% (231km) in good condition.
Roadside friction	▪ Roadside friction refers to activities that take place on and alongside the road which affect the travel performance and level of service of a road. ▪ 31% (1,680km) of the paved road network has no roadside friction, 62% (3,360km) has moderate roadside friction and 8% (409km) has severe roadside friction.
Potholes	▪ 6% (342km) of roads are in a moderate to severe condition due to the occurrence of potholes. Isolated potholes occur on a further 16% (892km) of the network and 77% (4,217km) have no potholes. Table 5-1 below details the roads heavily affected by potholes and the total length of the affected sections.

Table 4-1: List of roads heavily affected by potholes

Link ID	Link name	Length (km)		
		Moderate	Warning	Severe
A001_Link02	Mukono - Lugazi	3.3	0.0	0.0
A001_Link03	Lugazi - Njeru	4.0	0.0	0.0
A001_Link08	Nakalama - Bugiri	1.0	0.0	0.0
A004_Link01	Masaka roundabout - Kyotera	18.0	5.2	0.0
A004_Link02	Kyotera - Mutukula	25.8	5.0	0.0
A005_Link02	Bujuuko - Mityana	1.6	0.0	0.0
A005_Link03	Mityana - Naama - Myanzi	15.8	0.0	0.0
A005_Link04	Myanzi - Kiganda	9.7	1.0	0.0
A005_Link05	Kiganda-Kitenga	10.0	4.0	0.0
A005_Link06	Kitenga-Mubende-Lusalira	13.5	0.0	0.0
A005_Link09	Kyegegwa - Kakabala - Kyenjojo	17.3	2.0	0.0
A005_Link15	Kikorongo - Bwera - Mpondwe	22.1	6.0	0.0
A006_Link12	Gulu - Atiak	1.0	0.0	0.0
A006_Link13	Atiak - Nimule (Uganda/Sudan border	1.6	0.0	0.0
A008_Link01	Karuma - Olwiyo	4.0	1.0	0.0
A008_Link02	Olwiyo - Packwach	8.4	23.0	0.0
A008_Link03	Packwach - Nebbi	14.0	17.5	0.0
A008_Link04	Nebbi - Eruba	7.0	5.0	0.0
A009_Link01	Kampala - Nansana	1.0	0.0	0.0
A009_Link03	Busunju - Kiboga - Lwamata	2.0	0.0	0.0
A010_Link01	Nsambya-Kabalagala-Gaba	2.0	0.0	0.0
B100_Link02	Gayaza - Kalagi	1.0	2.0	0.0
B150_Link02	Bwizibwera - Ibanda	2.0	0.0	0.0
B151_Link01	Mbarara - Ishanyu	2.0	0.0	0.0
B151_Link02	Ishanyu - Kabwohe	3.1	0.0	0.0
B300_Link06	Ariamoi - Moroto	1.4	0.0	0.0
C005_Link01	Salaama Road	4.0	4.1	0.0
C154_Link01	Matugga - Semuto - Kapeka	15.0	19.0	0.0
C157_Link01	Entebbe-Nakiwogo	3.4	0.0	0.0
C158_Link01	Nakiwogo Statehouse road	1.5	0.0	0.0
C170_Link01	Namboole Access Road	1.0	0.0	0.0
C214_Link02	Lumbugu - Kyotera	5.1	1.0	0.0
C231_Link01	Lumbugu - Rakai	2.5	1.0	0.0
C511_Link01	Kasese - Kilembe mines	1.0	3.7	0.0
C540_Link01	Fortportal-Kichwamba-Karugutu	6.0	0.0	0.0
C540_Link02	Karugutu-Bundibugyo - Lamia	1.0	0.0	0.0
C684_Link01	Gulu Airport road	3.1	0.0	0.0
C923_Link01	Arapai - Railway Station	1.0	0.0	0.0

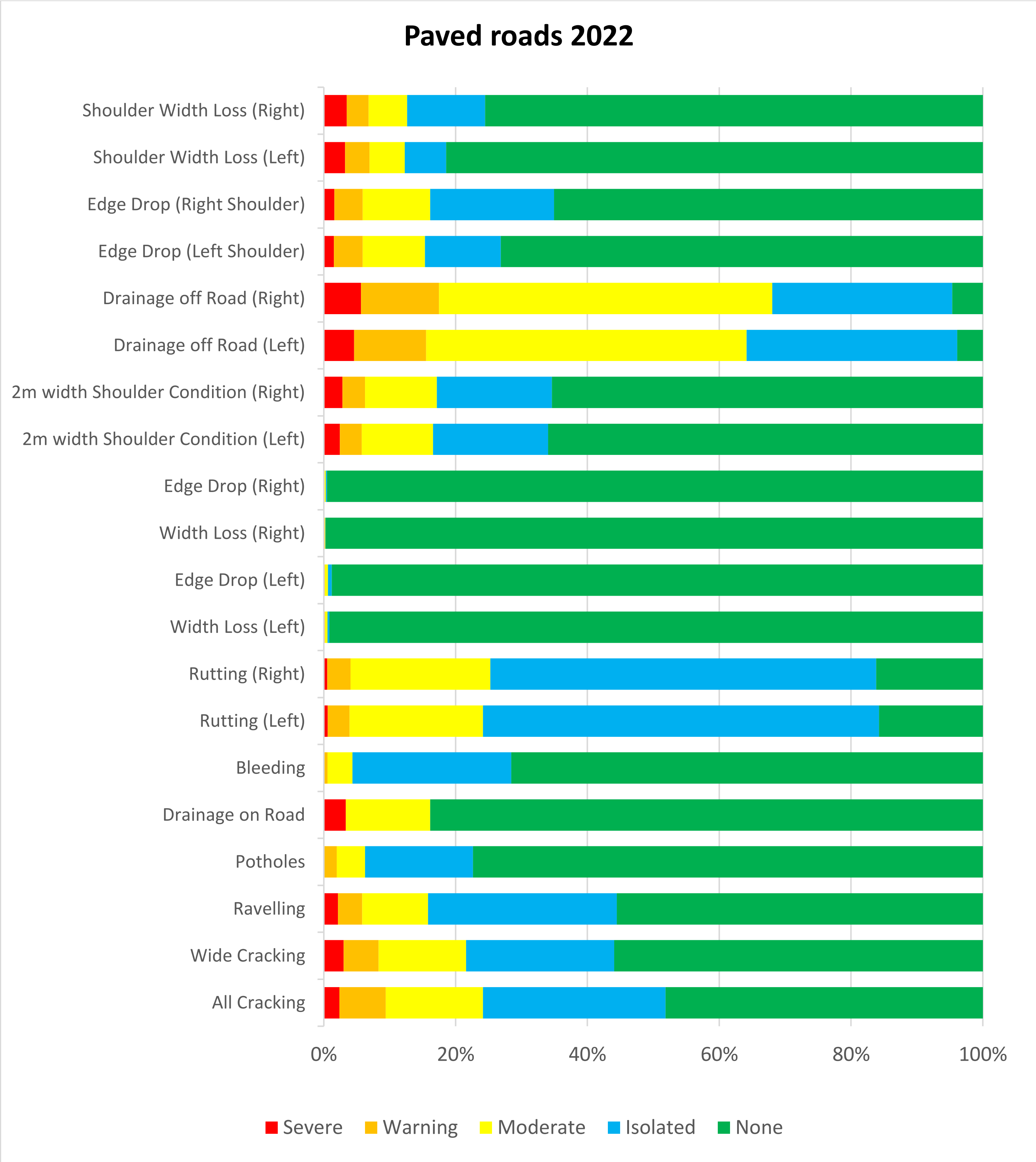
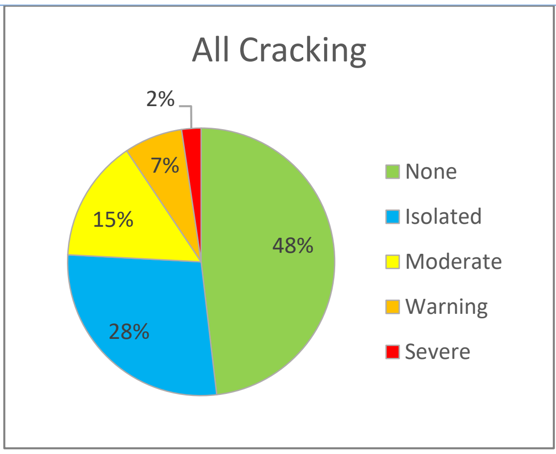
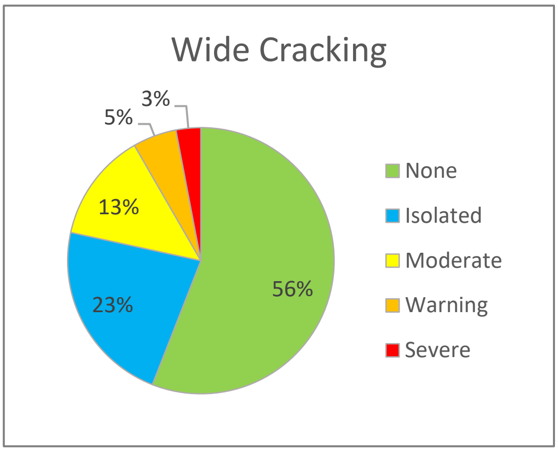
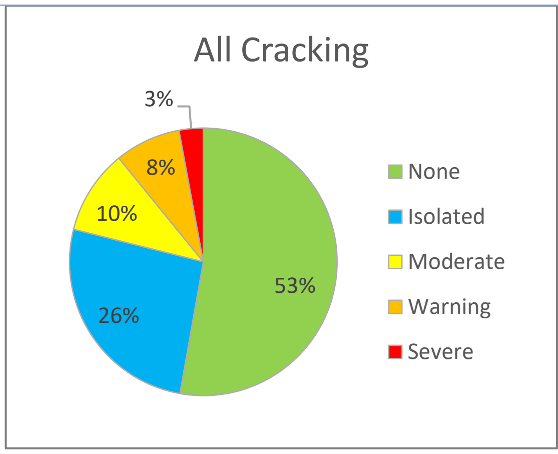


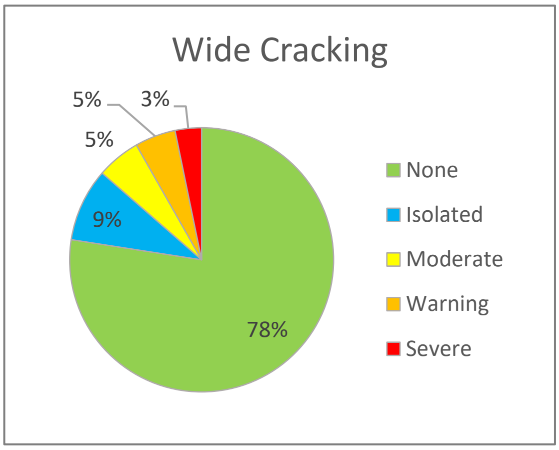
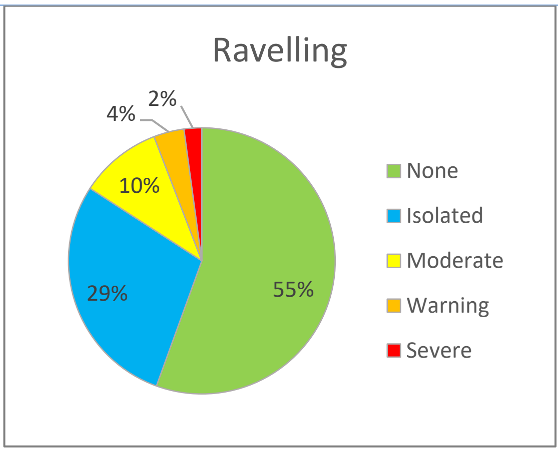
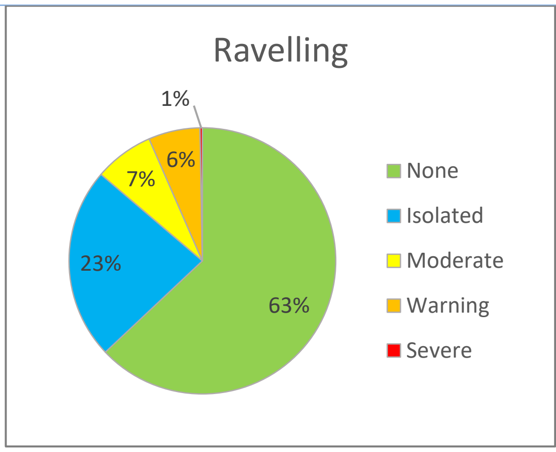
Figure 4-1: Measured values and distress ratings of paved roads, 2022

Generally, the paved road network is in good to fair condition. However, from a pavement management perspective, the roads appear to be structurally sound but an increase in cracking, ravelling, potholes, and bleeding indicates significant pavement deterioration. A preventive maintenance regime of timely patching, crack sealing, diluted emulsions and resurfacings will contribute significantly to delaying the structural deterioration by preventing water ingress into the pavement structures.

4.3 Comparison of the condition of Paved roads

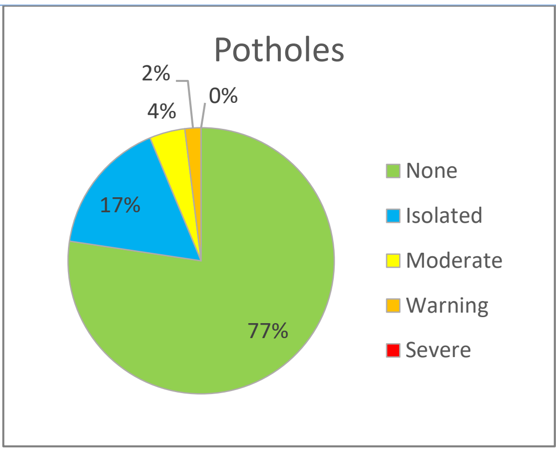
<table>
<tr><th>Distress</th><th>2022</th><th>2021</th></tr>
<tr><td rowspan="2">Cracking</td><td>

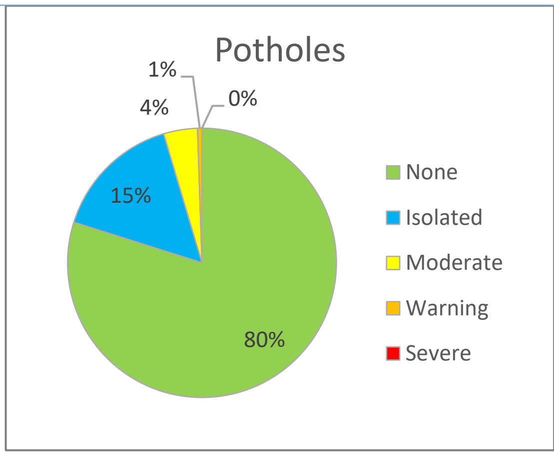

</td><td>


</td></tr>
<tr><td colspan="2">There has been a reduction in the length of paved roads with no all cracking (from 53% in 2021 to 48% in 2022) and no wide cracking (from 78% in 2021 to 56% in 2022). This indicates a deterioration in the condition and immediate crack sealing is required to extend the longevity of the road asset and reduce the long-term maintenance costs.</td></tr>
<tr><td>Ravelling</td><td>
</td><td>
</td></tr>
</table>

There is also a reduction in the length of paved roads with no ravelling from 63% in 2021 to 55% in 2022. The percentage length with isolated ravelling has increased from 23% in 2021 to 29% in 2022. These indicate a deterioration in performance. An intervention is required since ravelling reduces the skid resistance of a pavement and exposes the underlying layers to deterioration.

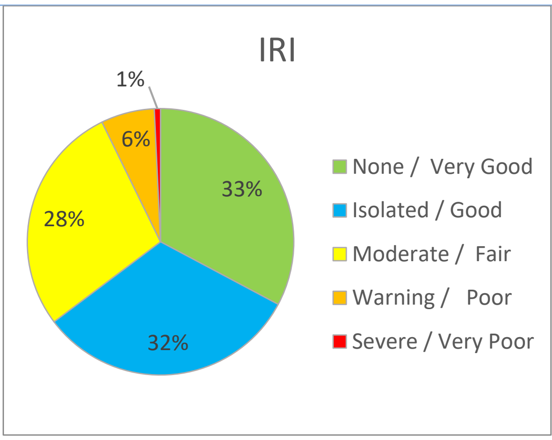
Potholes

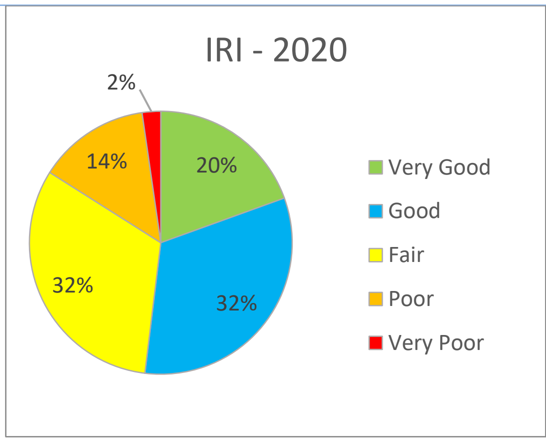




Basing on potholes, the condition of paved roads has declined from 80% of the length having no potholes in 2021 to 77% in 2022.

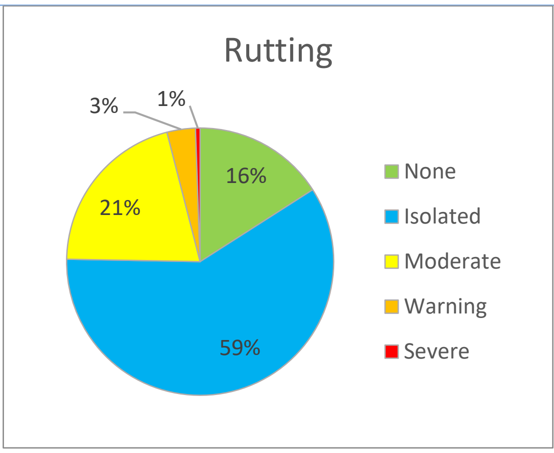
Roughness (measured)

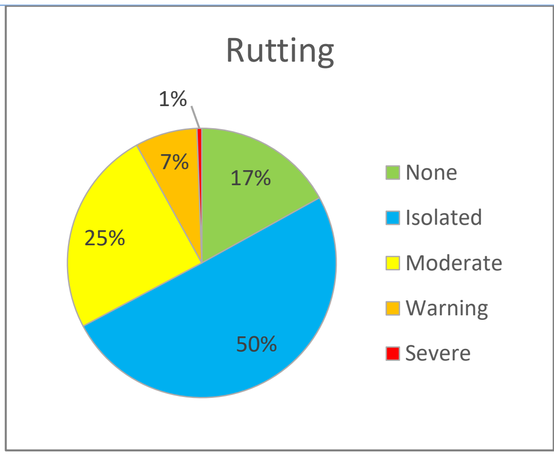


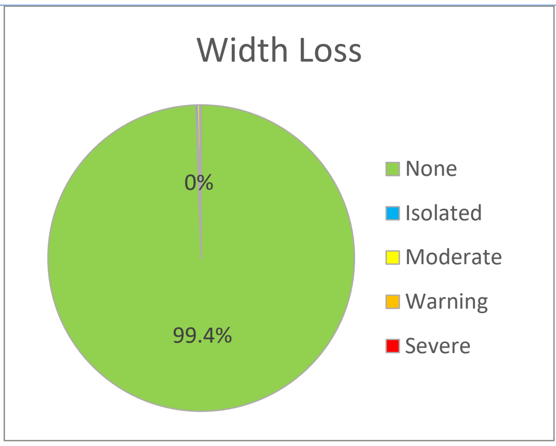
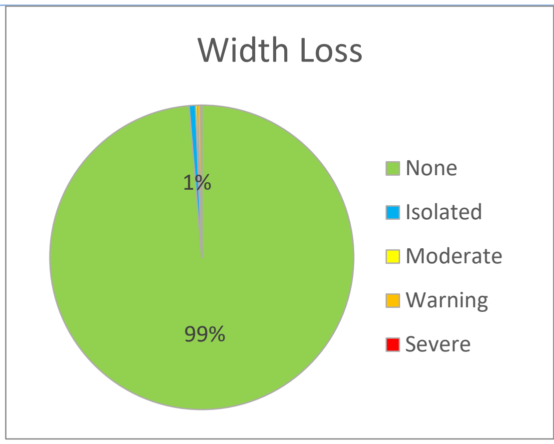
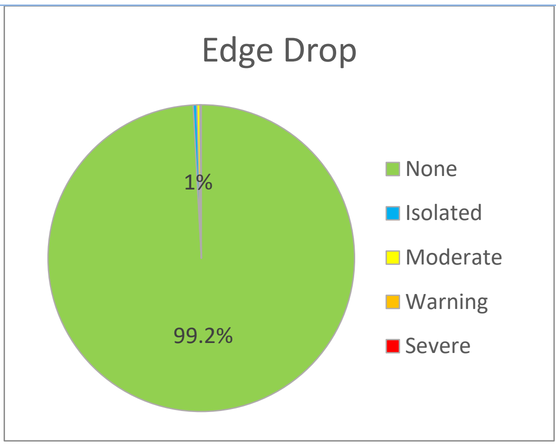
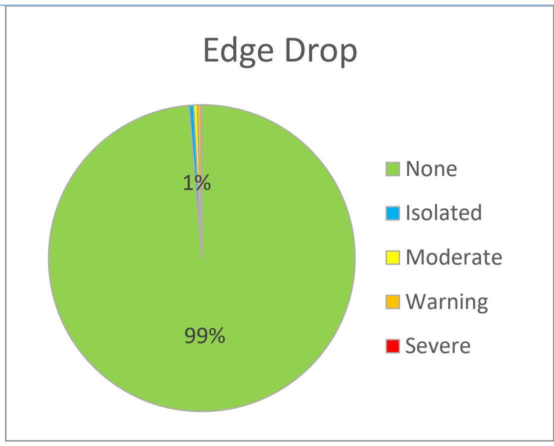
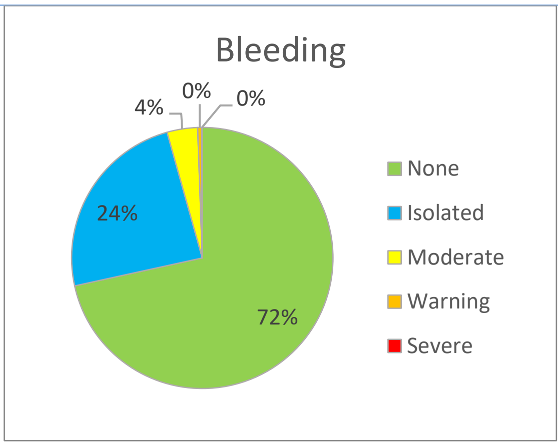
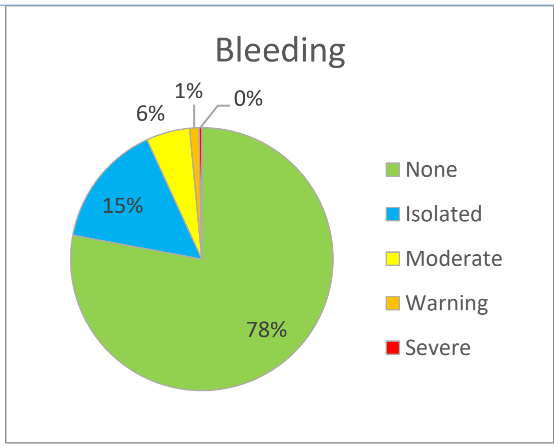


IRI surveys indicate an improvement in the paved road network from 52% in 2020 to 65% in 2022 in very good and good condition. The percentage in fair condition has also reduced from 32% in 2020 to 28% in 2022.

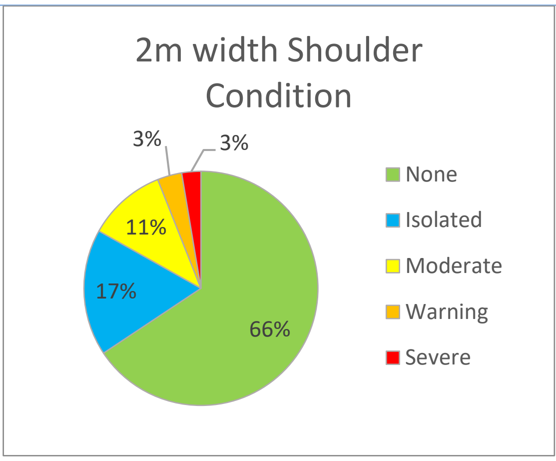
Rutting

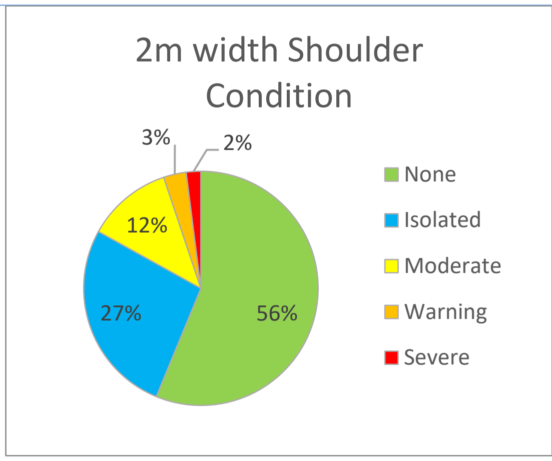




	There is an improvement in the performance with 75% in 2022 having none or isolated problems from 67% in 2020.
Width Loss	  No change in the extent of width loss on the paved network
Edge Drops	  No change in the extent of edge drop on the paved network
Bleeding	  There has been a reduction in the condition of road with no bleeding from 78% in 2021 to 72% n 2022. The percentage length with isolated bleeding has also increased from 15% in 2021 to 24% in 2022. This also indicates a deterioration in the condition of the paved roads.

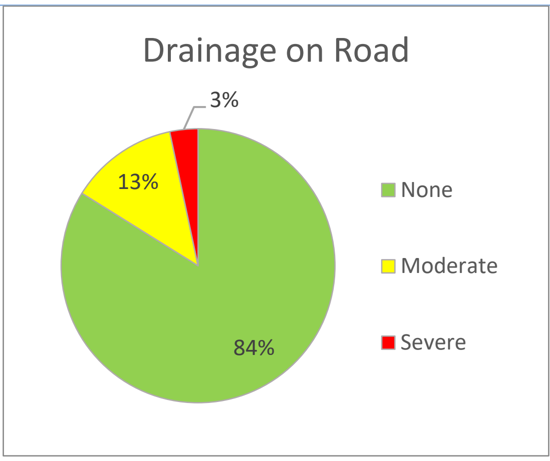
Shoulder condition

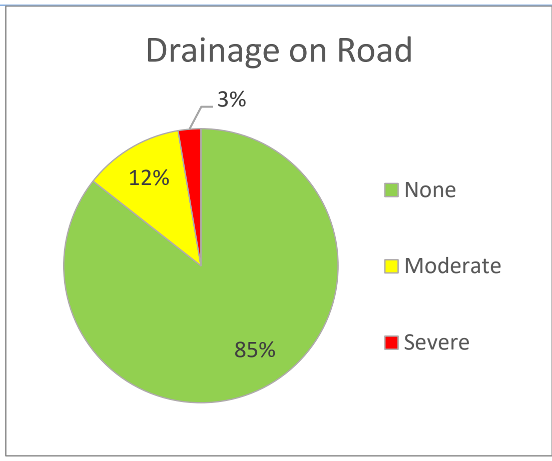




There has been an improvement in the shoulder condition of paved roads from 56% having no defect in 2021 to 66% in 2022. The percentage length with isolated problems has also reduced from 27% in 2021 to 17% in 2022.

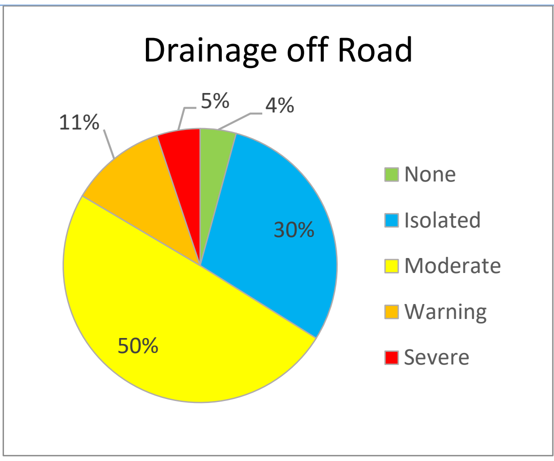
Drainage on road

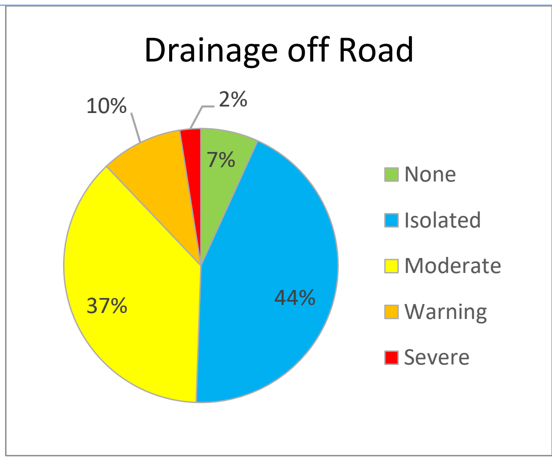




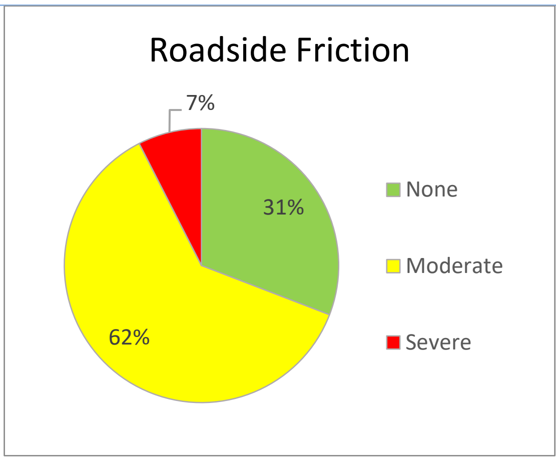
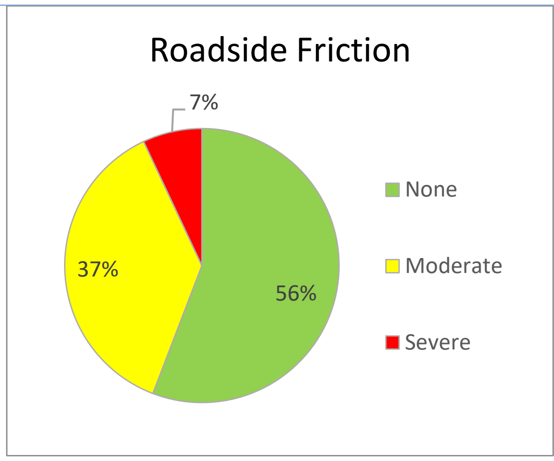
No significant change in the drainage on the road for the paved network

Drainage off road





Generally, there has been a reduction in the condition of drainage off the road with the percentage length of paved roads with none to isolated problems reducing from 53% in 2021 to 34% in 2022.

Roadside friction		
	Roadside friction has intensified on the network with 56% having no friction in 2021 to 31% in 2022.	

4.4 Average VCI for Paved roads

The average paved road VCI based on the 2022 data is 79% which categorises the road network in the “good” condition category. In 2021, the average paved road VCI was 83% indicating a decline of 4%. The graph below shows the comparison in VCI over the years.

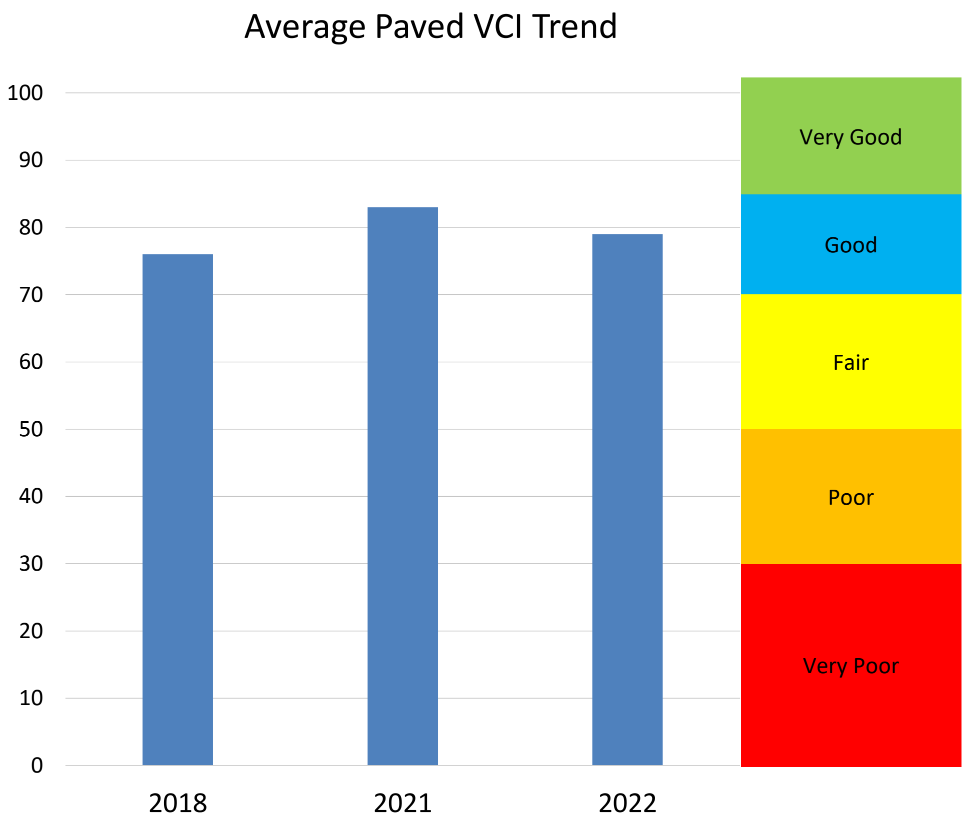


Figure 4-2: Average Paved VCI comparison

4.5 Condition distribution graphs for VCI of Paved Roads

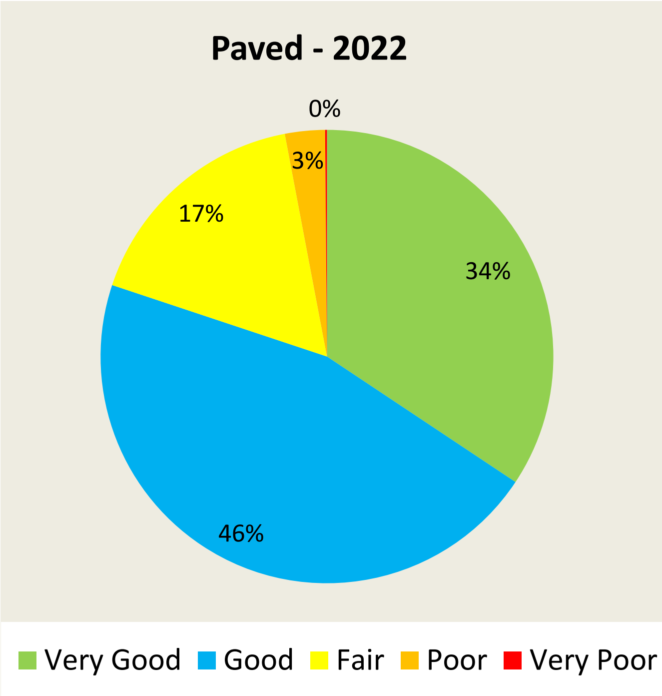


Figure 4-3: Paved VCI for 2022

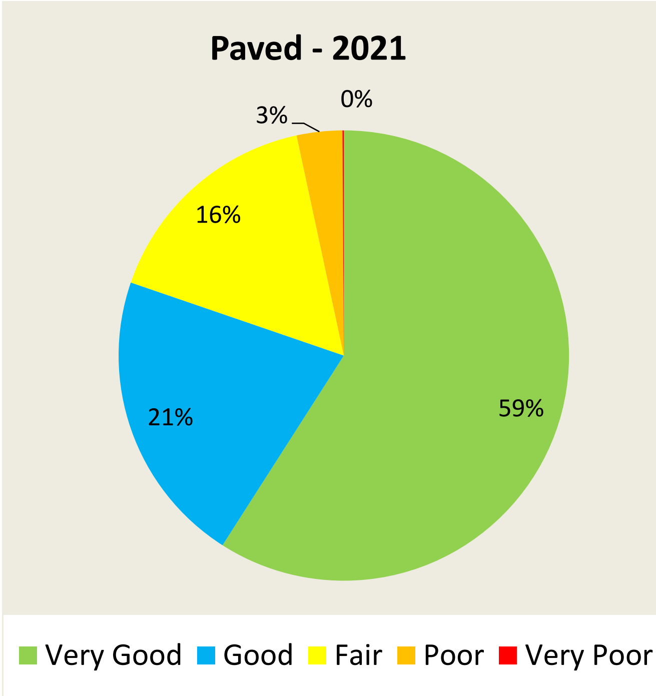


Figure 4-4: Paved VCI for 2021

Figures 4-2 and 4-3 above show that the percentage of roads in very poor, poor and moderate condition has generally remained constant. The roads in very good condition have reduced from 59% in 2021 to 46% in 2022 and those in good condition have increased from 21% in 2021 to 46% in 2022. This shows a decline in the performance of the roads in very condition. However, the total roads in good and very good condition have remained constant at 80%.

Generally, the paved road network is in good condition with only 3% in poor and very poor condition, indicating a need for rehabilitation. The roads in fair condition indicate a need for preventive maintenance (see Appendix 2 for details). Timely maintenance is recommended to preserve the current asset value.

4.6 Condition trend for Paved roads

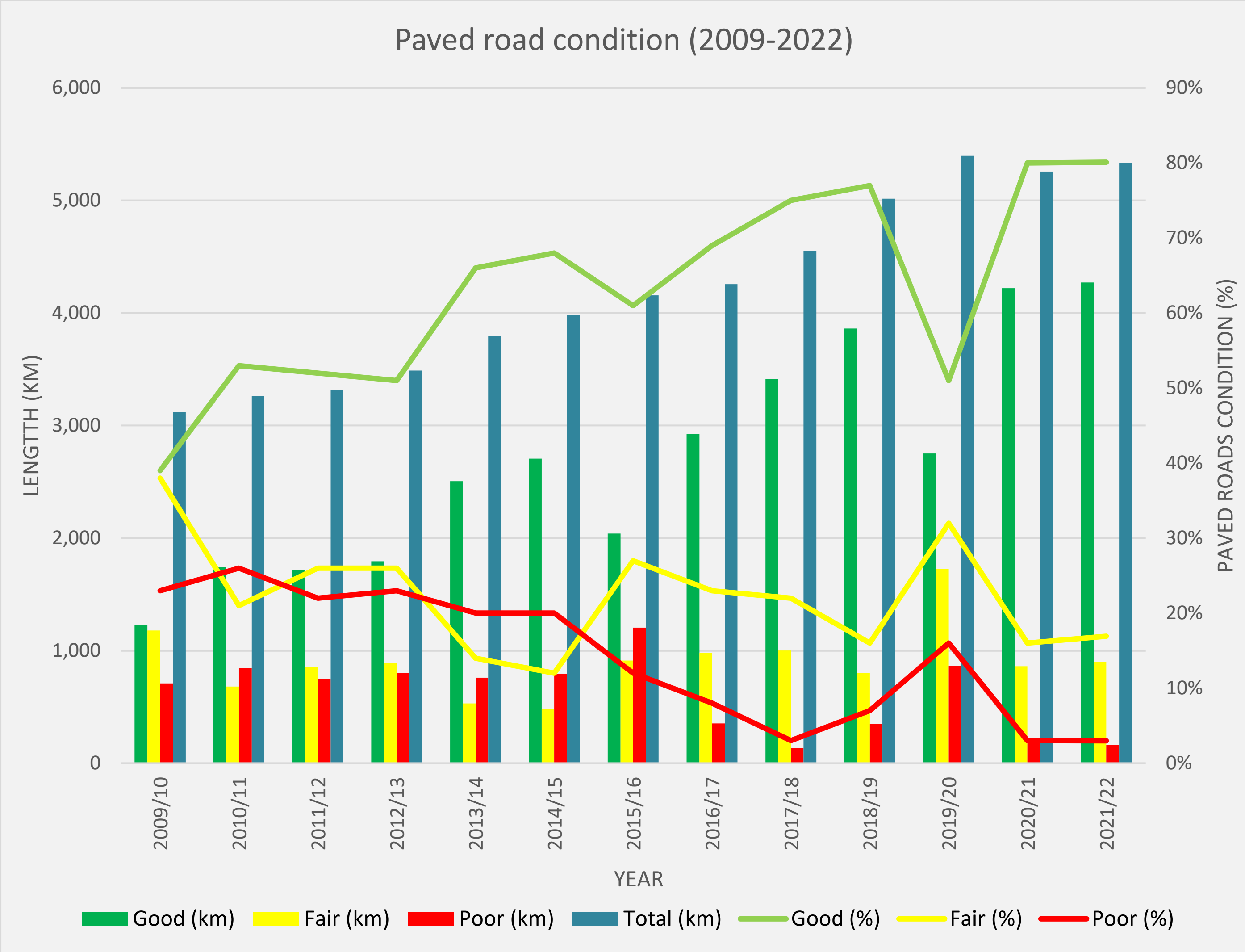


Figure 4-5: Condition trend for the paved roads from 2009/10 to 2021/22

Table 4-2: Condition of paved roads from 2009/10 to 2021/22

YEAR	Paved Roads Condition (km)				Paved Roads Condition (%)		
	Good	Fair	Poor	Total	Good	Fair	Poor
2009/10	1,230	1,180	709	3,119	39%	38%	23%
2010/11	1,742	680	843	3,264	53%	21%	26%
2011/12	1,717	856	744	3,317	52%	26%	22%
2012/13	1,794	893	803	3,490	51%	26%	23%
2013/14	2,505	531	759	3,795	66%	14%	20%
2014/15	2707	478	796	3981	68%	12%	20%
2015/16	2040	913	1204	4157	61%	27%	12%
2016/17	2924	979	354	4257	69%	23%	8%
2017/18	3413	1001	136	4551	75%	22%	3%
2018/19	3862	802	351	5015	77%	16%	7%
2019/20	2753	1727	864	5398	51%	32%	16%
2020/21	4220	861	177	5258	80%	16%	3%
2021/22	4272	902	159	5333	80%	17%	3%

4.7 Performance of the paved road network per station

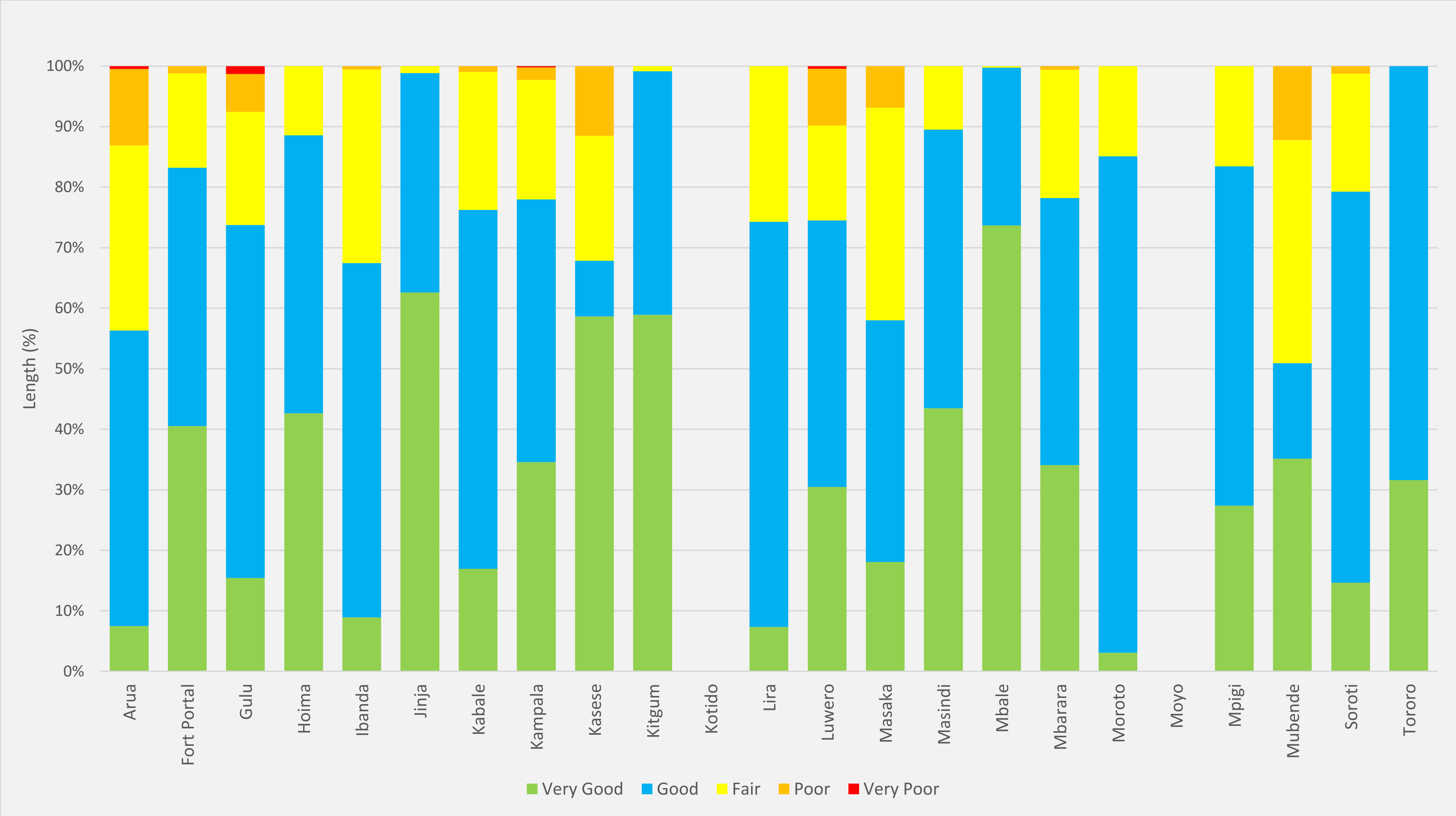


Figure 4-6: VCI rating of the paved road network per station

Figure 4-5 above indicates the performance of the paved road network by station. Most stations maintain the road network in good and very good condition. Mbale (74%) and Jinja (63%) stations have the highest percentage of roads in very good condition while Gulu (1%) station has the highest percentage of roads in very poor conditions (see Appendix 3 for details).

5. Unpaved Road Network Condition

- *The average unpaved road VCI based on the 2022 data is 62% which categorises the road network in the "fair" condition category.*
- *The gravel thickness ratings indicate the need for urgent re-gravelling on 19% (2,327km). 1% (88km) has no gravel at all and 19% (2,239km) has gravel less than 50mm.*
- *The roughness ratings on the unpaved network indicate severe to moderate condition states on 39% (4,892km) of the road network.*
- *Roads with rutting in a moderate to severe condition account for almost 45% (5,619km) of the network.*
- *22% (2,747km) of all roads have severe to moderate corrugations.*
- *Erosion is very common with almost 30% (3,766km) of roads showing the occurrence of severe to moderate erosion.*
- *Drainage condition and formation level are poor and very poor on 10% (1,165km) of the network.*
- *% Of Roads in very poor and poor condition has decreased from 23% in 2020 to 11% in 2022.*

5.1 Background to Unpaved VCI

The assessment ratings are utilised to calculate a Visual Condition Index (VCI) similar to the VCI of paved roads. This unpaved VCI is calculated for each unpaved road segment that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average unpaved network VCI is calculated representing the average condition of the unpaved road network or any sub-network. This average network VCI is weighted by length and is used to describe/investigate the average condition of the road network/sub-network.
- The VCI is furthermore used to categorise the condition of an unpaved road segment, the unpaved road network or sub-network into one of five condition categories. The condition categories adopted are:

- Very Good = 86% to 100%
- Good = 71% to 85%
- Fair = 51% to 70%
- Poor = 31% to 50%
- Very Poor = 0% to 30%

5.2 Visual assessment of Unpaved Road network

Unpaved visual assessment data was collected in 2020 and the PMS was updated with this data.

The assessment data expresses the condition and characteristics of the wearing course material and the functional condition of the gravel roads as a set of rating scores, rating these as 1 (very good or non-existent), 2 (slight, good or isolated occurrence), 3 (discernible, fair or moderate occurrence), 4 (warning, poor or general occurrence) and 5 (severe, very poor or extensive occurrence).

These ratings can indirectly be related to the proportion of the unpaved road network with none, isolated, moderate, warning and severe problems.

According to the visual assessments the following conclusions can be made with regard to the unpaved roads (refer to

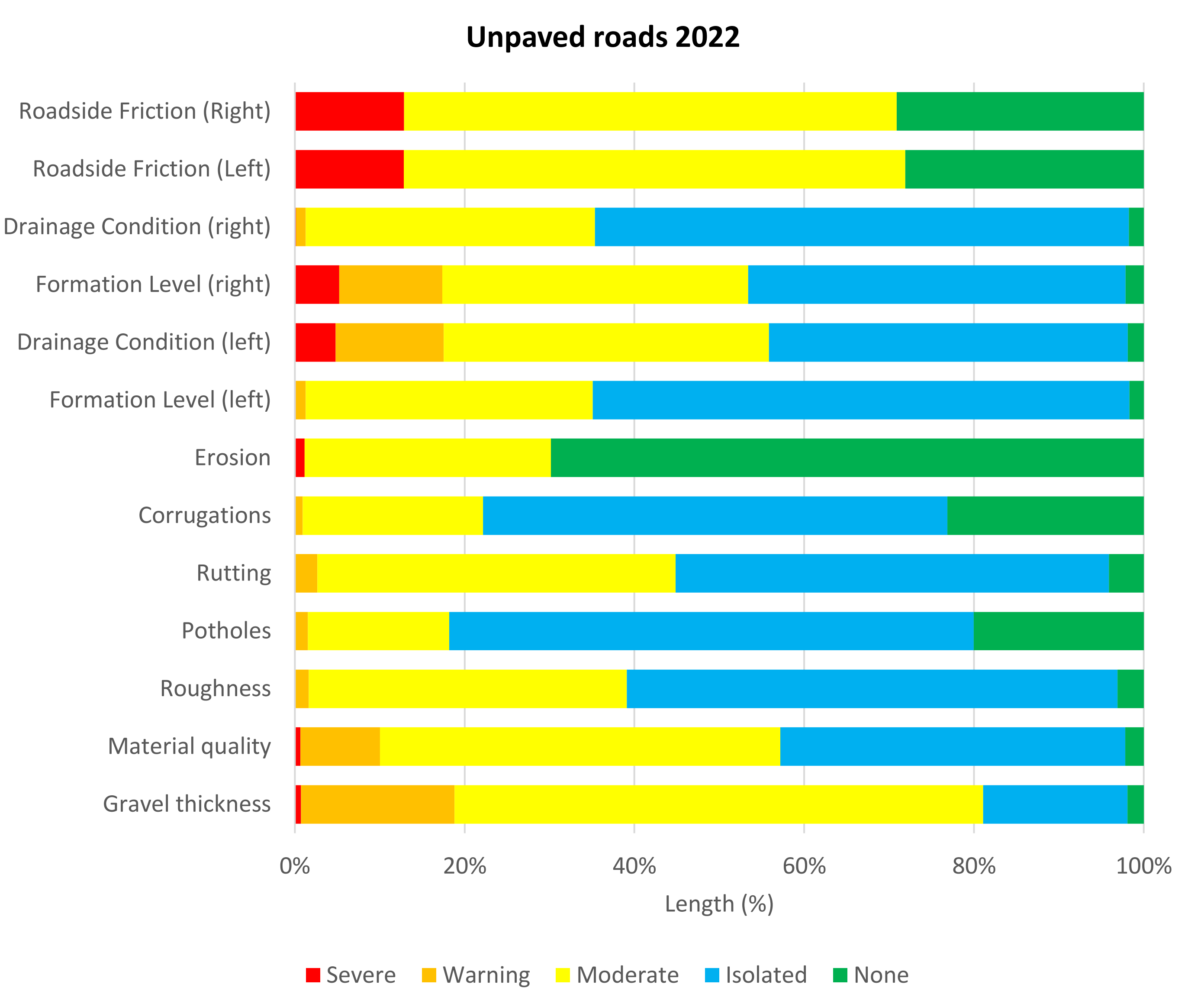


Figure 5-1: Distress, characteristic and condition ratings of unpaved roads, 2022

Gravel thickness	The gravel thickness ratings indicate the need for re-gravelling on 19% (2,327km) of the unpaved road network, with 1% (88km) having no gravel, another 18% (2,239km) having less than 50mm gravel wearing course and severe exposure of the subgrade. A further 63% (7,966km) is in the moderate condition state, with subgrade exposed in local areas (average of 50 to 100mm gravel thickness).
Material quality	1% (83km) of roads having severe material quality exhibiting a poorly distributed range of particle sizes and/or zero or excessive plasticity, and 9% (1,159km) of roads were rated with warning material quality (poor particle size distribution). A further 48% (6,088km) exhibits cracking, loose material or stones clearly visible, thus rated as material quality of moderate state.

Roughness	▪ The roughness ratings on the unpaved network indicate 39% (4,892km) of the road network to be in severe to moderate condition. This will result in high road user costs. 58% (7,357km) have isolated problems with the ride smooth/fair and comfortable
Potholes	▪ The high occurrence of potholes especially in severe to moderate states on 17% (2,253km) and 62% (7,903km) in the isolated state is indicative of inadequate maintenance.
Rutting	▪ Roads with rutting in a moderate to severe condition account for close to 45% (5,619km) of the network. These roads require blading and re-gravelling.
Corrugations	▪ 22% (2,747km) of all roads have severe to moderate corrugations. A further 56% (7,024km) has isolated corrugations. Although these roads require blading, there seems to be a need to investigate whether material properties contribute to the formation of corrugations.
Erosion	▪ Erosion was rated moderate to severe on 30% (3,766km) of the road network. There is a need to repair erosion gulleys more regularly.
Drain condition and formation level	▪ Drainage condition and formation level indicate the ability of roads to drain water away, transversely underneath and longitudinally next to them. Roads with severe or warning formation levels would act as water channels, this is the case for 10% (1,154km) of the unpaved road network. 35% (4,396km) of the unpaved roads have moderate formation levels indicating that the drainage is level with ground level. ▪ Furthermore, the drainage condition is warning to severe for 9% (1,164km) of the network with moderate drainage conditions existing on 36% (4,572km) of the network.
Roadside friction	▪ 28% (3,561km) of the unpaved road network has no roadside friction, 59% (7,496km) has moderate roadside friction and 13% (1,589km) has severe roadside friction.

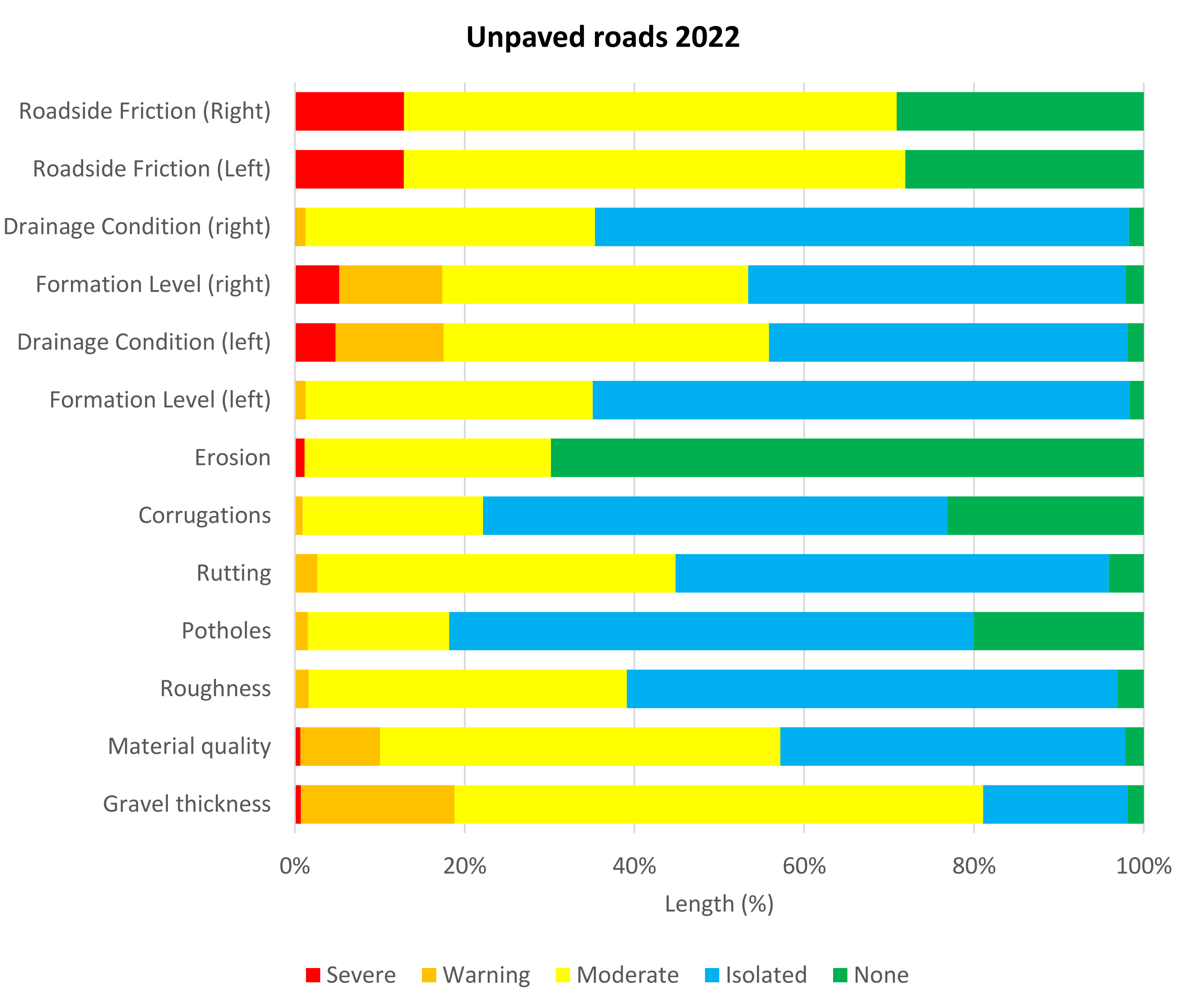
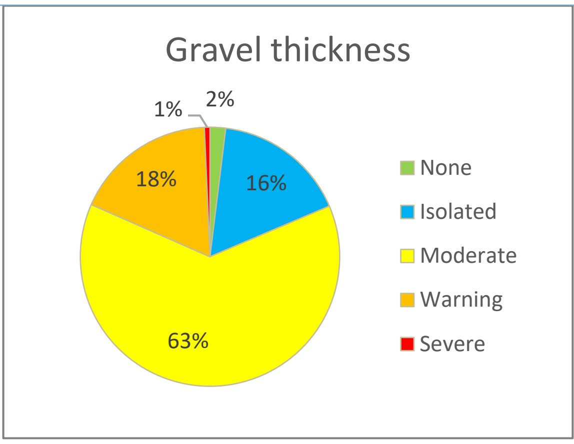
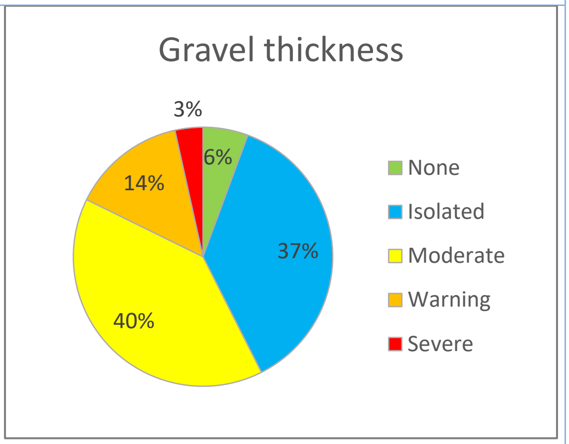
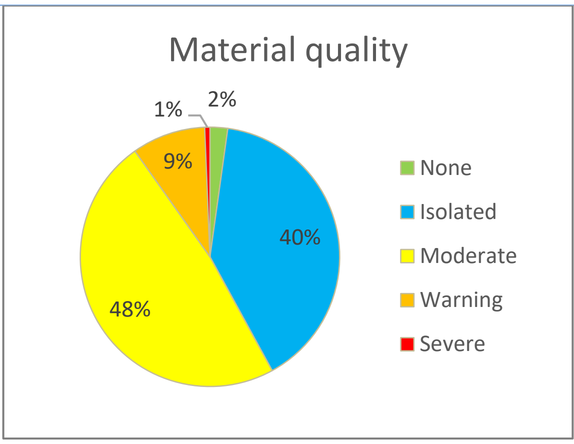
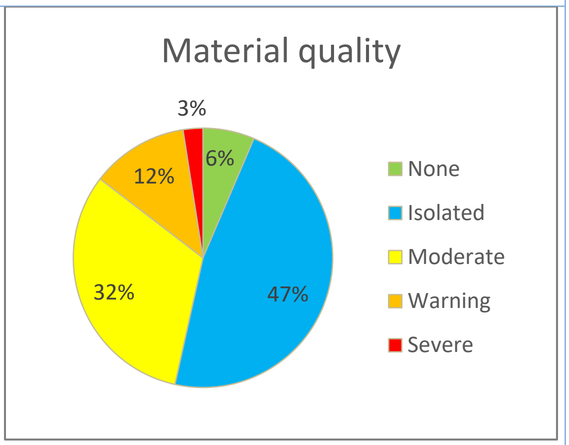


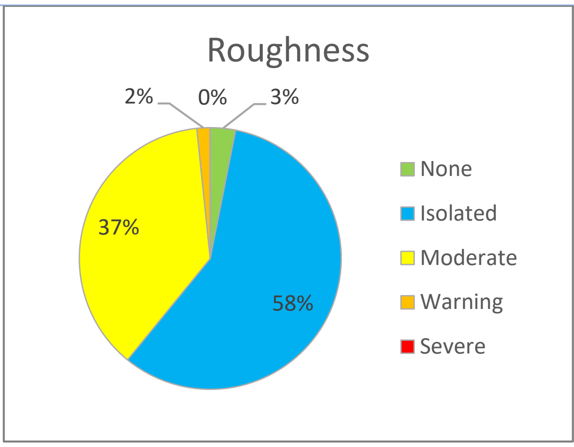
Figure 5-1: Distress, characteristic and condition ratings of unpaved roads, 2022

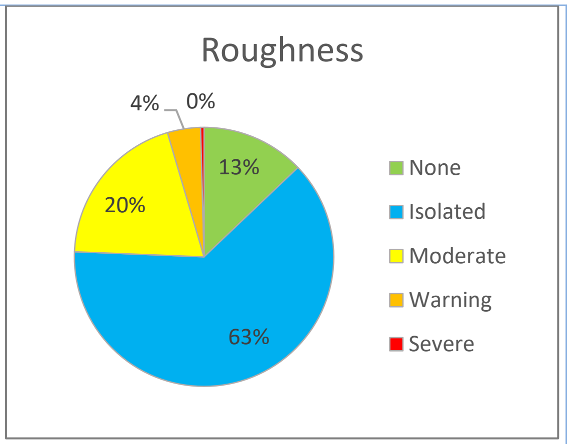
In summary, the unpaved network is in fair condition. From a road user's perspective, material quality and possibly non-sufficient maintenance regimes contribute to the development of rough roads with potholes, rutting and corrugations. From an unpaved roads management perspective, gravel thickness problems, low formation levels and poor drainage conditions result in erosion, wetting of the gravel layers and subgrade which result in potholes and rutting. A well-planned re-gravelling programme should be supported by an effective maintenance regime of blading (to maintain cambers and limit roughness), cleaning of side drains, mitre drains and cross drainage.

5.3 Comparison of the condition of Unpaved roads

<table>
<tr><th>Distress</th><th>2022</th><th>2021</th></tr>
<tr><td rowspan="2">Gravel thickness</td><td>
</td><td>
</td></tr>
<tr><td colspan="2">Generally, there has been a decline in the performance in regards to material quality. The percentage length with no defect has reduced from 6% in 2021 to 2% in 2022. That with isolated occurrence of the defect has also decreased from 37% in 2021 to 16% in 2022 and the percentage with moderate occurrence has increased from 40% in 2021 to 63% in 2022. Timely regravelling is advised to deter quick deterioration of the unpaved network.</td></tr>
<tr><td rowspan="2">Material quality</td><td>
</td><td>
</td></tr>
<tr><td colspan="2">Generally, there is an improvement in the material quality with 15% in 2021 having warning to severe quality to 10% in 2022. However, there is a percentage reduction of unpaved roads with good material quality from 6% in 2021 to 2% in 2022. The percentage with moderate material quality has increased from 32% in 2021 to 48% in 2022.</td></tr>
</table>

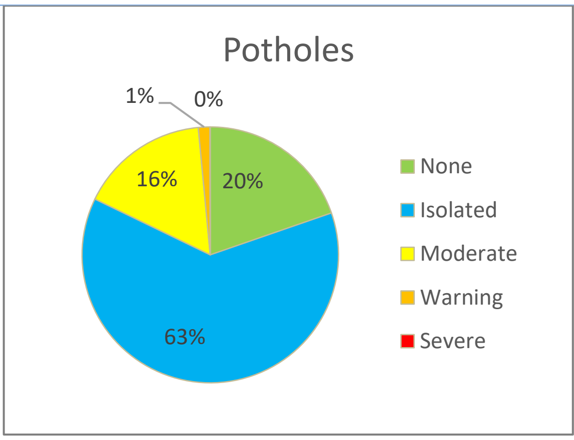
Roughness

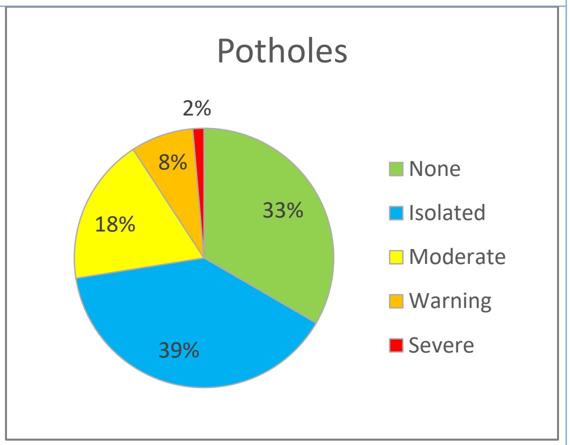




Basing on roughness, the condition of unpaved roads has declined from 13% of the length having a smooth ride quality in 2021 to 3% in 2022. The percentage length with isolated roughness has also reduced from 63% in 2021 to 58% in 2022. The length with moderate roughness has increased from 20% in 2021 to 37% in 2022. Regular and timely maintenance is required to keep the road network in a good motorable condition.

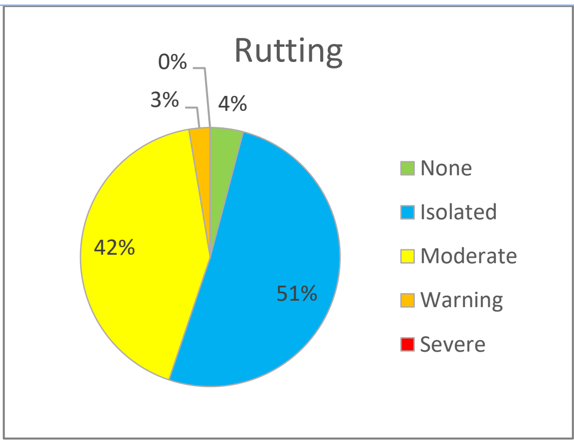
Potholes

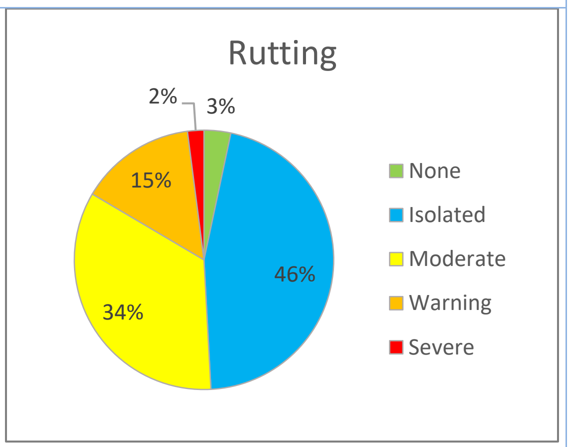




Generally, there is an improvement of performance with regards to potholes on the unpaved road network. The percentage in warning to severe state has reduced from 10% in 2021 to 1% in 2022, however the percentage length with no potholes has reduced from 33% in 2021 to 20% in 2022.

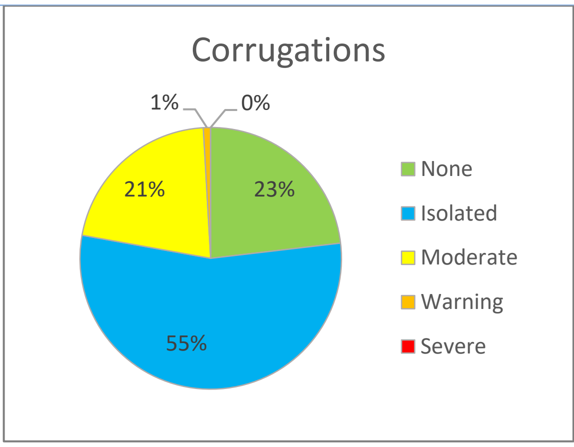
Rutting

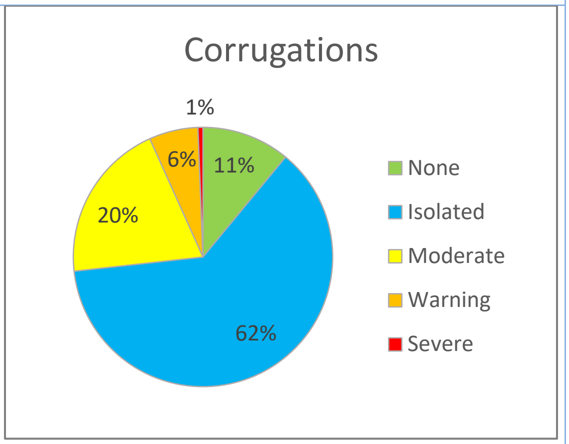




There has been an improvement in the performance of unpaved roads with regards to rutting with 3% having severe and warning occurrences of rutting in 2022 from 17% in 2021. The percentage with none and isolated rutting has also increased to 55% in 2022 from 49% in 2021.

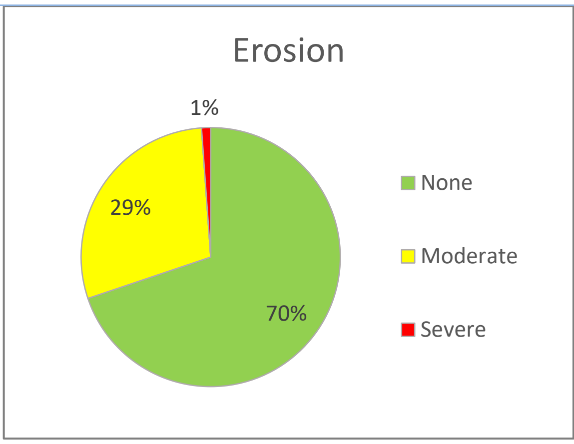
Corrugations

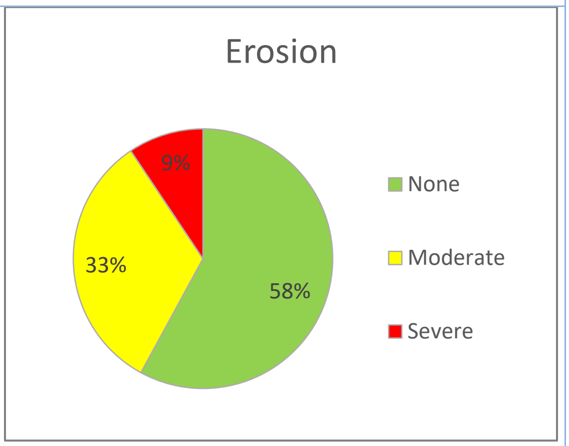




The percentage length with no corrugations has increased from 11% in 2021 to 24% in 2022. The percentage with warning and severe corrugations has also reduced from 7% in 2021 to 1% in 2022. This points to an improvement in the performance with regards to corrugations.

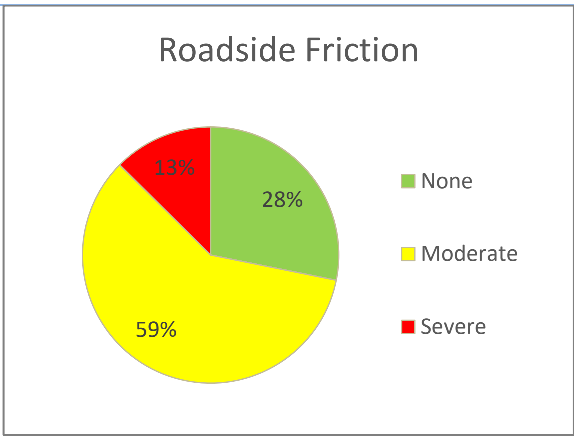
Erosion

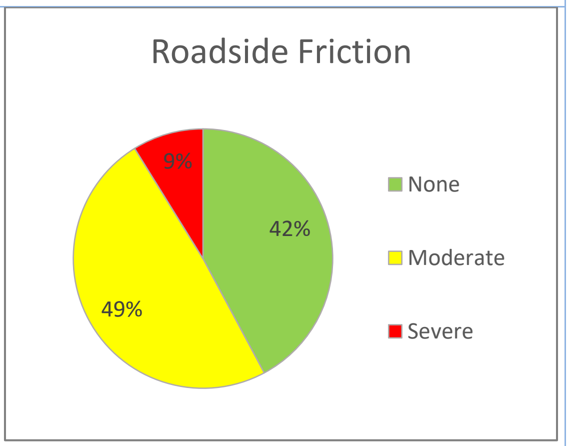




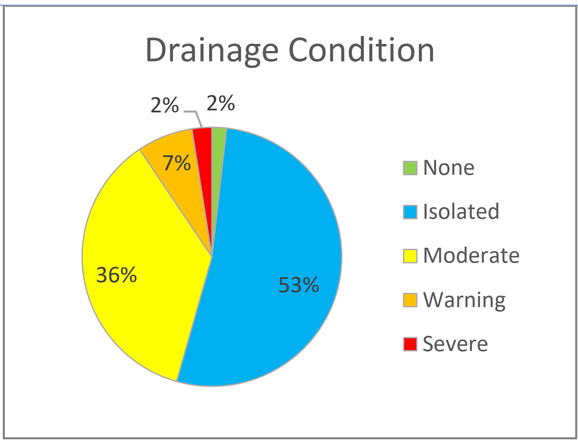
There has also been an improvement with regards to erosion with 1% having severe erosion in 2022 down from 9% in 2021. The percentage with no erosion has increased from 58% in 2021 to 70% in 2022.

Roadside friction

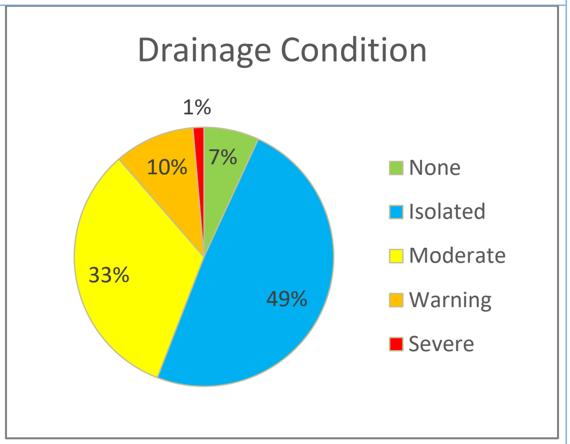




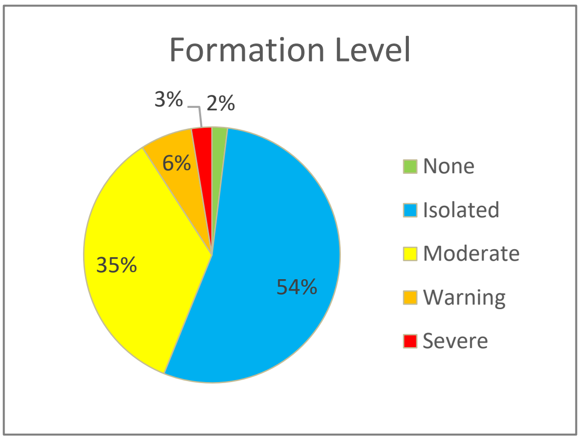
<table>
<tr><td></td><td colspan="2">Roadside friction has intensified on the network with 42% having no friction in 2021 to 28% in 2022. The percentage with severe roadside friction has also increased from 9% in 2021 to 13% in 2022.</td></tr>
<tr><td rowspan="3">Drain condition and formation level</td><td>



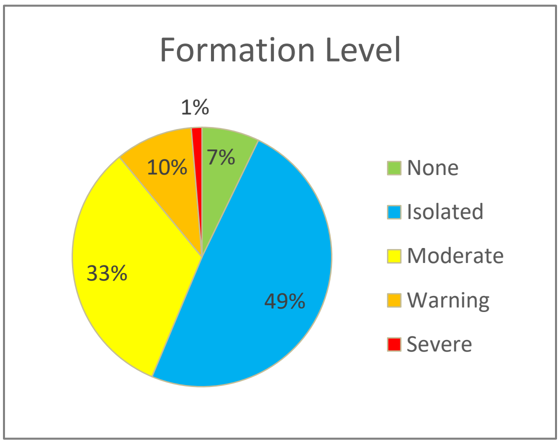
</td><td>



</td></tr>
<tr><td>



</td><td>



</td></tr>
<tr><td colspan="2">Generally, there is a slight improvement with 9% in 2022 having warning to severe drain condition and formation level from 11% in 2021. However, there is a decline in length with no condition defect from 7% in 2021 to 2% in 2022.</td></tr>
</table>

5.4 Average VCI for Unpaved roads

The average unpaved road VCI based on the 2022 data is 62% which categorises the road network in the “fair” condition category, this has remained constant compared to the average VCI calculated on the 2021 visual assessment data. The graph below shows the trend of VCI for the unpaved road network.

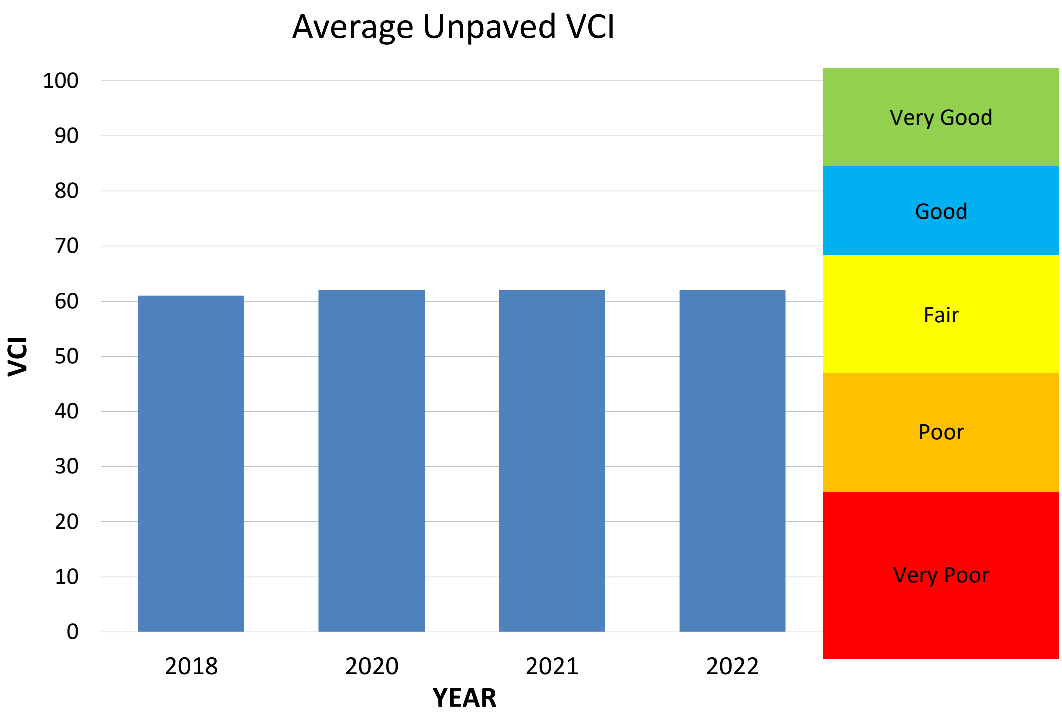
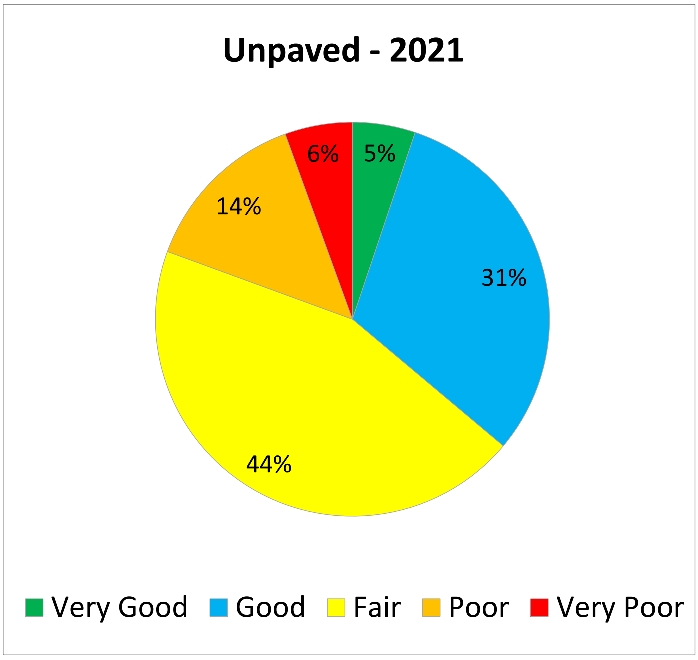


Figure 5-2: Average Unpaved VCI comparison (2018-2022)

5.5 Condition distribution graphs for the VCI of Unpaved roads



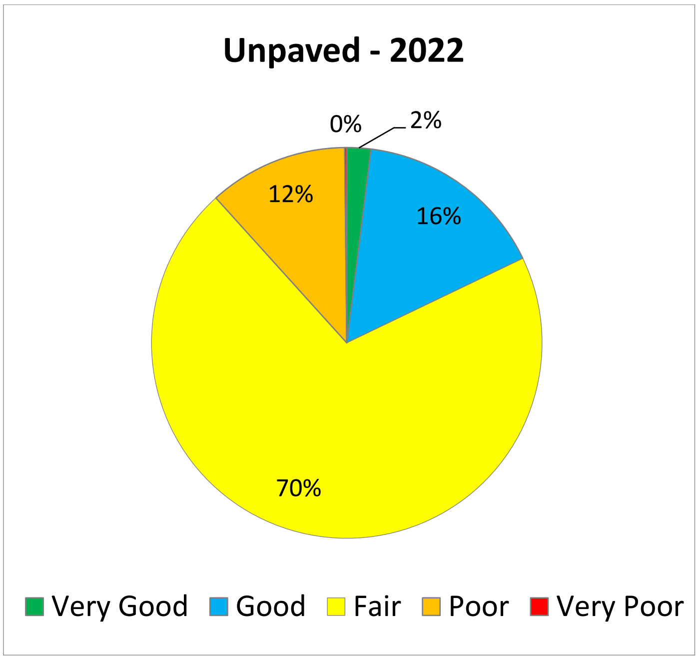


Figure 5-3: Unpaved VCI distribution for 2021 **Figure 5-4: Unpaved VCI distribution for 2022**

The figures show that the percentage of roads in very poor and poor condition has reduced from 20% in 2021 to 12% in 2022. The roads in fair condition have increased from 44% in 2021 to 70% in 2022. The condition of roads in good and very good condition has decreased from 36% in 2021 to 18% in 2022.

5.6 Condition trends for Unpaved roads

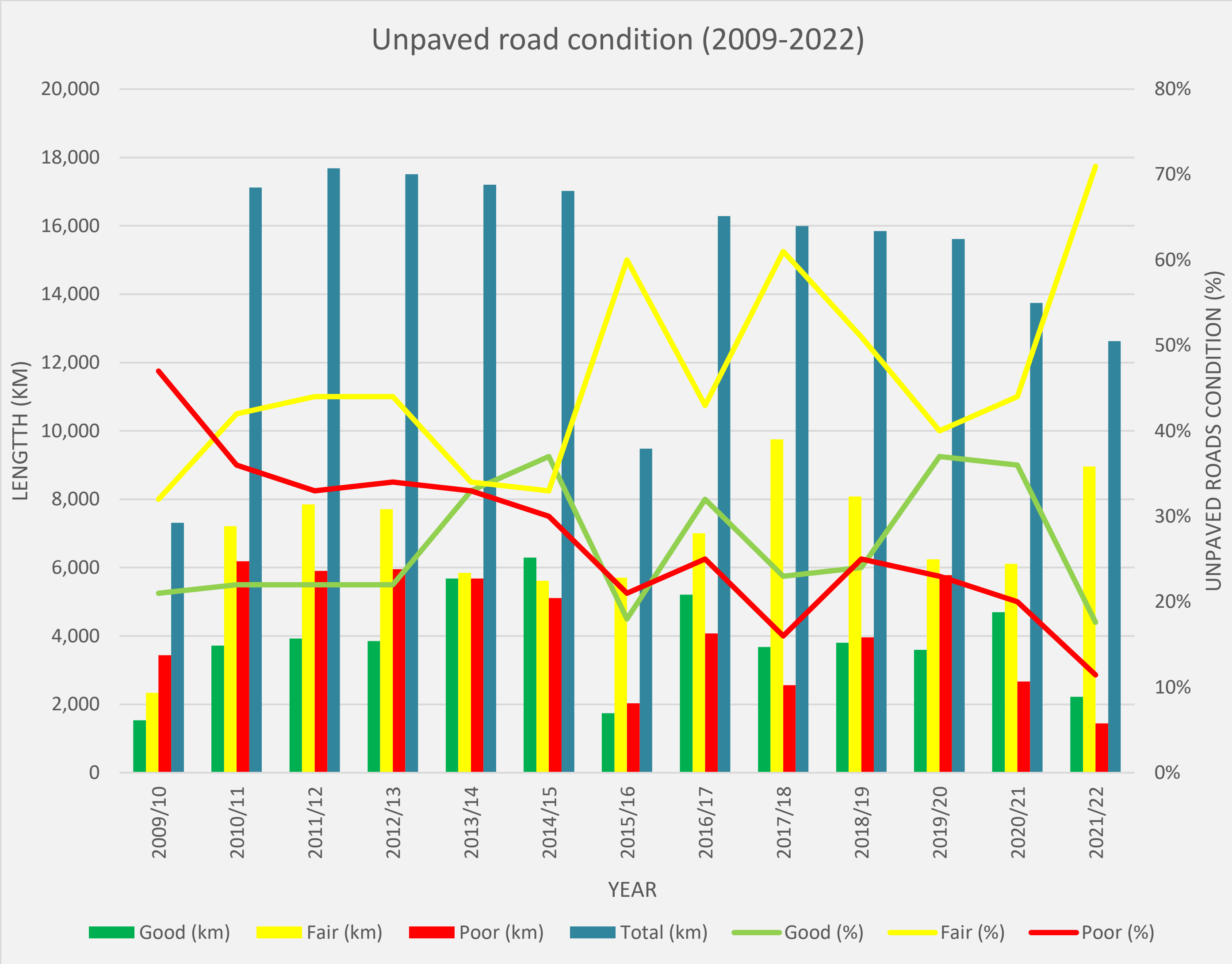


Figure 5-5: Condition trend for the unpaved roads from 2009/10 to 2021/22

Table 5-1: Condition of unpaved roads from 2009/10 to 2021/22

YEAR	Unpaved Roads Condition (km)				Unpaved Roads Condition (%)		
	Good	Fair	Poor	Total	Good	Fair	Poor
2009/10	1,535	2,340	3,436	7,311	21%	32%	47%
2010/11	3,719	7,215	6186	17,120	22%	42%	36%
2011/12	3,926	7,853	5,904	17,683	22%	44%	33%
2012/13	3,852	7,705	5,954	17,510	22%	44%	34%
2013/14	5,678	5,850	5,677	17,205	33%	34%	33%
2014/15	6297	5616	5106	17019	37%	33%	30%
2015/16	1737	5707	2033	9477	18%	60%	21%
2016/17	5212	7003	4072	16287	32%	43%	25%
2017/18	3678	9755	2558	15993	23%	61%	16%
2018/19	3802	8079	3960	15841	24%	51%	25%
2019/20	3591	6245	5776	15612	37%	40%	23%
2020/21	4695	6111	2669	13745	36%	44%	20%
2021/22	2223	8961	1445	12629	18%	71%	11%

5.7 Performance of the unpaved road network per station, 2022

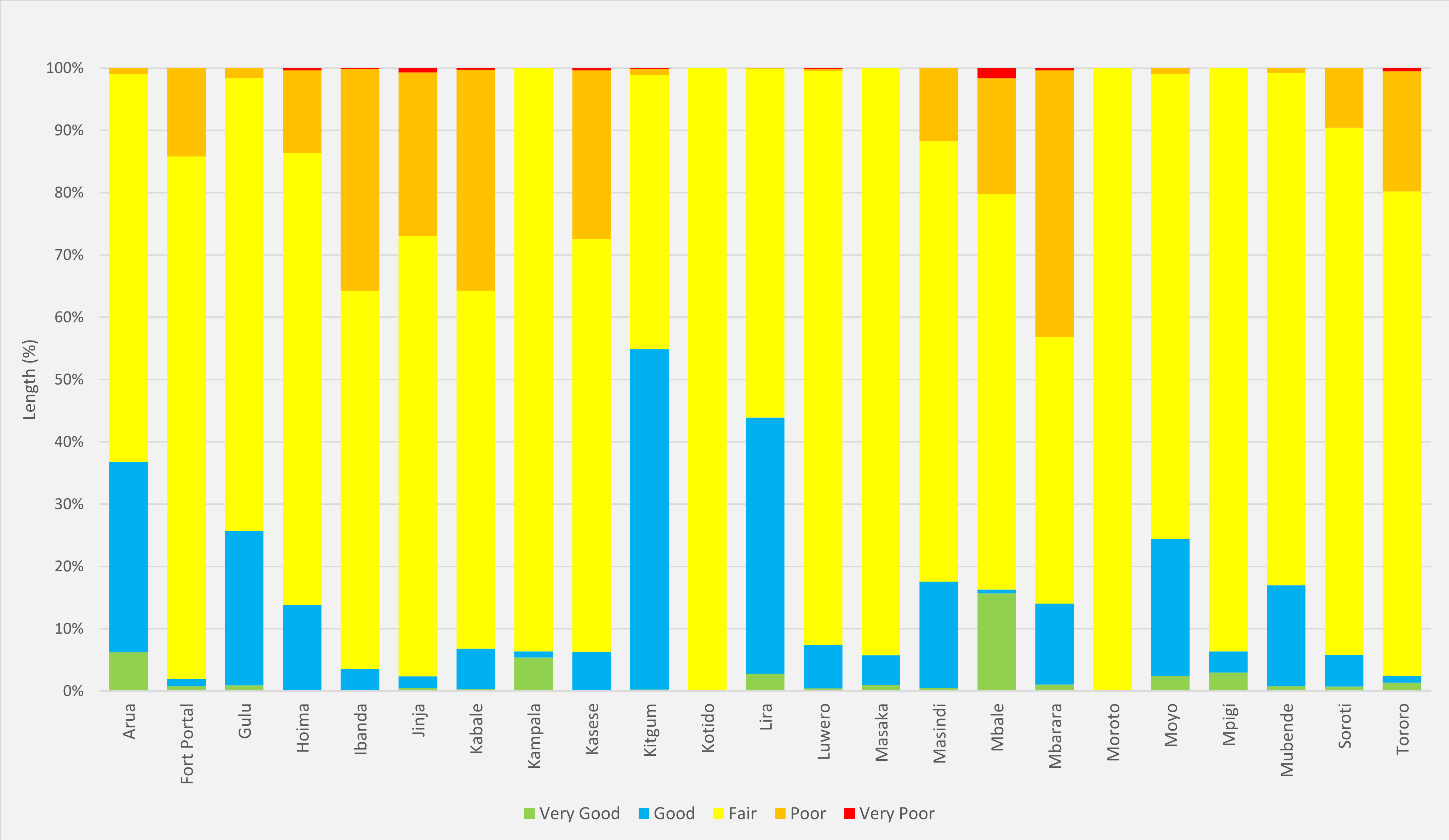


Figure 5-6: Condition rating of the unpaved road network per station

Figure 5-6 above indicates the performance of the unpaved road network by station. Most stations maintain the road network in good to fair condition. Mbale (16%), Arua (6%) and Kampala (5%) stations have the highest percentage of roads in very good condition while Mbale (2%), Jinja (1%) and Tororo (1%) stations have the highest percentage of roads in very poor conditions (see Appendix 6 for details).

6. Condition of Bridges and Major Culverts

Summary:

- *71% (236No.) of bridges are in good condition*
- *6% (25No.) bridges have the roadway in critical condition*
- *85% (275No.) of the major culverts are in good condition*
- *10% (34No.) major culverts have the roadway in critical condition*

Visual inspections were carried out to ascertain the inventory of the structure, identify defects, the stage and extent of the defect and the maintenance recommendations. Inspections were carried out by qualified bridge inspectors and must consisted of thorough examination of all the components of the structure. For bridges, the components assessed were; Waterways (if applicable), Foundations, Substructure (Abutments and Piers), Bearings, Superstructure, Non-structural elements (joints, drainage, parapets etc.) and Approaches. For Major Culverts, the components assessed were; Waterways, Inlets and Outlets, Substructure and Superstructure (Cell Invert, Cell Roof Slab, Cell walls), Precast Pipe Culvert, Corrugated Steel Pipe and Roadway.

A numerical scale (0-9) is used to rate the condition of both the main components and subcomponents. These ratings are based on the current prevailing condition of the component and are used to calculate an overall rating of the structure. The Overall Rate calculated is then classified in 5 condition categories, i.e. critical, poor, marginal, satisfactory, good, as follows;

Table 6-1: Condition categories of the Overall Rating

Calculated Overall Rate	Condition Category
0 and 1	Critical
2 and 3	Poor
4 and 5	Marginal
6	Satisfactory
7 to 9	Good

6.1 Condition of Bridges

Inventory and condition assessment of bridges and major culverts was done by the Data Collection Consultants in the Northern, Southern and Western region for the year 2022. A delay in the contract signing of Lot 1 (Central region) and issuance of Call off order 2 for Lot 4 (Eastern/North Eastern region) rendered it impossible to have a full network assessment by the time of preparing this report. For purposes of this analysis, historical assessment data was used for the unassessed structures. Based on this, the following conclusions can be made with regards to the condition of Bridges on the national network.

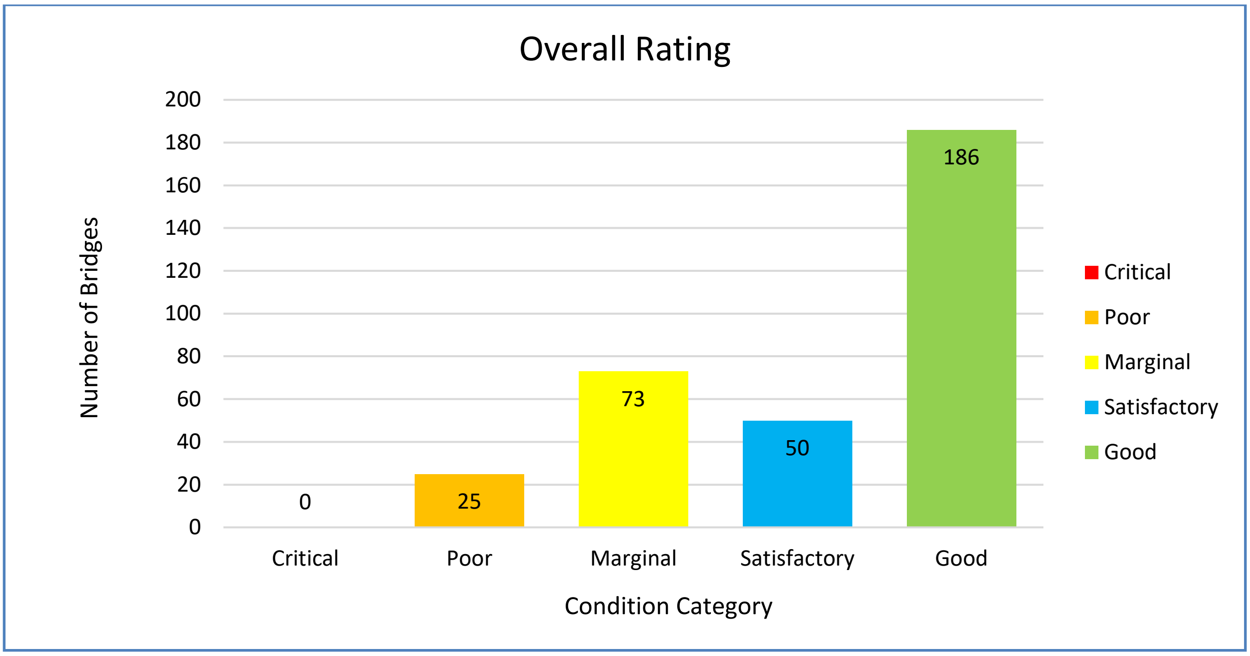


Figure 6-1: Overall rating of bridges

From Figure 6-1 above, it can be observed that the highest number of bridges is in good condition (186No.) and 50No bridges are in satisfactory condition. The bridges in marginal condition are significant (73No.) and timely maintenance is advised to prevent them from quickly deteriorating to a poor condition. Bridges in poor condition are the least (25No.) and urgent intervention is required. No bridge is in critical condition. Appendix 9 details the bridges in poor condition.

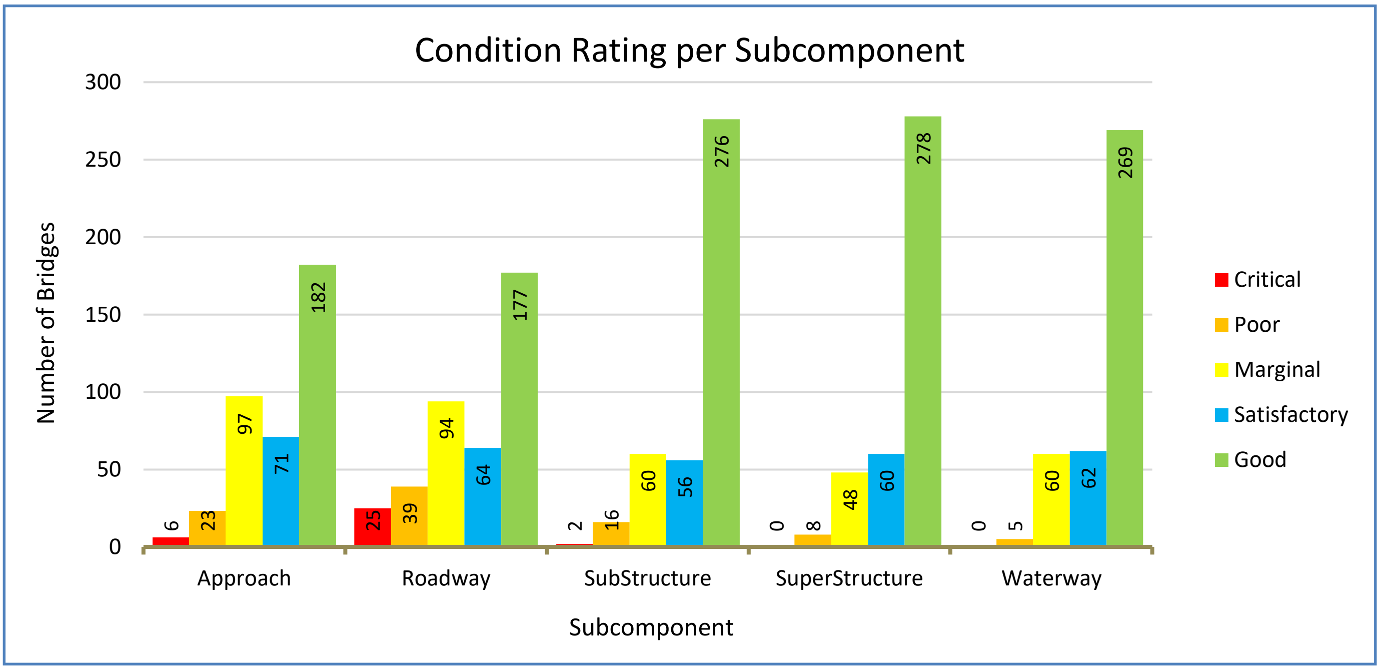


Figure 6-2: Condition rating of bridges per subcomponent

From the figure above, it is observed that;

- The majority of bridges have subcomponents in good condition.
- An appreciable number of subcomponents have a marginal rating and timely maintenance is required.
- Subcomponents in poor and critical condition are minimal and displaying in only the approaches and roadway and urgent maintenance is required. Failure to do so possess a safety hazard and can lead to closure or collapse of a bridge.

6.2 Condition of Major Culverts

Inventory and condition surveys were carried out as stated in 6.1 above. Based on these visual assessments, the following conclusions can be made with regards to the condition of Major Culverts on the national network.

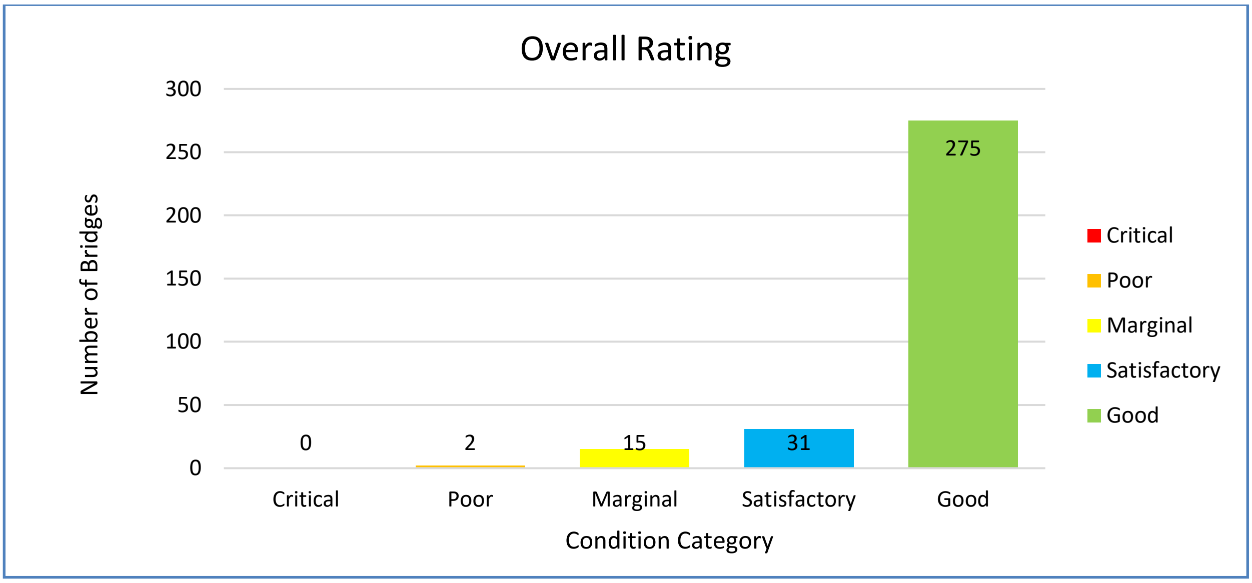


Figure 6-3: Overall rating of major culverts

From Figure 6-3 above, it can be observed that 275No. major culverts are in good condition. 31No. are satisfactory and 15No. are in marginal condition. Only 2No. major culverts are in poor and critical condition and immediate intervention is required (see Appendix 9 for details).

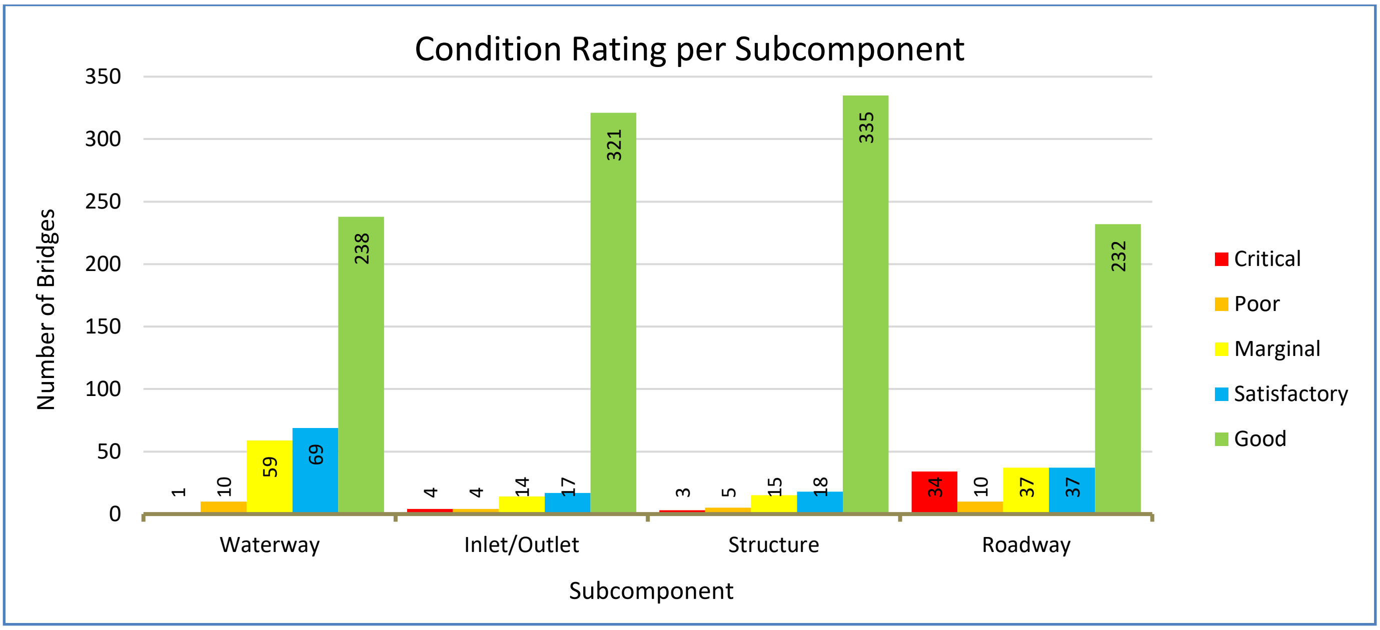


Figure 6-4: Condition rating of major culverts per subcomponent

From the figure above, it is observed that;

- The majority of major culverts have subcomponents in good condition.
- Subcomponents in poor and critical condition are minimal and displaying in mostly the roadway. More effort is required in timely maintenance of the roadway.

7. Conclusion

The UNRA road network's total length is 21,120km. Of these roads 5,745km (27%) are paved roads and 15,375km (73%) are unpaved roads. For these roads, there is condition data available for 18,032km (85%) of road, of which 5,675km (99%) is paved roads data and 12,357km (80%) is unpaved roads data.

The overall condition of the paved road network is good but shows a decline, while the overall condition of the unpaved network is fair and has remained constant.

On average, the condition of the structures on the national road network is good with a limited number in a poor state.

The situational analysis now sets the scene for the roads needs analysis.

8. Appendices

Appendix 1: Pictures of the UNRA network

Appendix 4: Traffic Summary, 2022

Appendix 2: Visual Condition Summary Paved roads, 2022

Appendix 3: Performance of the paved road network per station, 2022

		VCI rating (km)					VCI rating (%)				
Station	**Length surveyed**	**Very Good**	**Good**	**Fair**	**Poor**	**Very Poor**	**Very Good**	**Good**	**Fair**	**Poor**	**Very Poor**
Arua	213.711	16.00	104.37	65.34	27.00	1.00	7%	49%	31%	13%	0%
Fort Portal	363.606	147.43	155.18	56.67	4.33	0.00	41%	43%	16%	1%	0%
Gulu	389.284	60.16	226.99	72.77	24.37	5.00	15%	58%	19%	6%	1%
Hoima	324.033	138.15	148.83	37.05	0.00	0.00	43%	46%	11%	0%	0%
Ibanda	190.690	17.00	111.65	61.04	1.00	0.00	9%	59%	32%	1%	0%
Jinja	264.640	165.72	95.92	3.00	0.00	0.00	63%	36%	1%	0%	0%
Kabale	318.316	53.91	188.86	72.55	3.00	0.00	17%	59%	23%	1%	0%
Kampala	484.213	167.43	210.13	95.66	10.00	1.00	35%	43%	20%	2%	0%
Kasese	130.440	76.52	11.96	26.96	15.00	0.00	59%	9%	21%	11%	0%
Kitgum	119.572	70.45	48.12	1.00	0.00	0.00	59%	40%	1%	0%	0%
Kotido	0.000										
Lira	64.523	4.72	43.20	16.60	0.00	0.00	7%	67%	26%	0%	0%
Luwero	241.622	73.63	106.44	37.88	22.68	1.00	30%	44%	16%	9%	0%
Masaka	283.997	51.29	113.50	99.71	19.50	0.00	18%	40%	35%	7%	0%
Masindi	181.443	78.89	83.55	19.00	0.00	0.00	43%	46%	10%	0%	0%
Mbale	297.935	219.57	77.69	0.68	0.00	0.00	74%	26%	0%	0%	0%
Mbarara	343.868	117.24	151.71	72.92	2.00	0.00	34%	44%	21%	1%	0%
Moroto	162.725	5.00	133.51	24.21	0.00	0.00	3%	82%	15%	0%	0%
Moyo	0.000										
Mpigi	268.333	73.51	150.40	44.42	0.00	0.00	27%	56%	17%	0%	0%
Mubende	172.665	60.70	27.27	63.61	21.08	0.00	35%	16%	37%	12%	0%
Soroti	231.308	33.88	149.50	45.00	2.93	0.00	15%	65%	19%	1%	0%
Tororo	179.298	56.67	122.63	0.00	0.00	0.00	32%	68%	0%	0%	0%

Appendix 5: Visual Condition Summary Unpaved roads, 2022

Appendix 6: Performance of the unpaved road network per station, 2022

		VCI rating (km)					VCI rating (%)				
Station	**Length surveyed**	**Very Good**	**Good**	**Fair**	**Poor**	**Very Poor**	**Very Good**	**Good**	**Fair**	**Poor**	**Very Poor**
Arua	693.020	42.84	212.10	431.21	6.86	0.00	6%	31%	62%	1%	0%
Fort Portal	590.905	4.38	7.00	495.81	83.72	0.00	1%	1%	84%	14%	0%
Gulu	553.187	5.00	137.14	401.92	9.12	0.00	1%	25%	73%	2%	0%
Hoima	609.860	0.00	84.13	442.59	81.00	2.14	0%	14%	73%	13%	0%
Ibanda	637.746	0.00	22.49	387.22	227.04	1.00	0%	4%	61%	36%	0%
Jinja	888.465	3.91	17.03	628.12	233.19	6.22	0%	2%	71%	26%	1%
Kabale	789.581	2.04	51.55	453.85	280.05	2.09	0%	7%	57%	35%	0%
Kampala	496.656	26.71	4.73	465.22	0.00	0.00	5%	1%	94%	0%	0%
Kasese	288.639	0.00	18.19	191.19	78.26	1.00	0%	6%	66%	27%	0%
Kitgum	878.801	2.00	480.42	386.70	8.69	1.00	0%	55%	44%	1%	0%
Kotido	17.500	0.00	0.00	17.50	0.00	0.00	0%	0%	100%	0%	0%
Lira	794.968	22.00	327.01	444.96	1.00	0.00	3%	41%	56%	0%	0%
Luwero	828.108	3.51	57.31	763.28	3.00	1.00	0%	7%	92%	0%	0%
Masaka	630.442	6.00	30.02	594.42	0.00	0.00	1%	5%	94%	0%	0%
Masindi	209.564	1.00	35.78	148.17	24.62	0.00	0%	17%	71%	12%	0%
Mbale	491.761	77.07	3.00	312.01	91.59	8.08	16%	1%	63%	19%	2%
Mbarara	585.210	6.00	76.00	250.67	250.54	2.00	1%	13%	43%	43%	0%
Moroto	19.944	0.00	0.00	19.94	0.00	0.00	0%	0%	100%	0%	0%
Moyo	546.902	13.00	120.65	408.25	5.00	0.00	2%	22%	75%	1%	0%
Mpigi	398.155	11.84	13.40	372.92	0.00	0.00	3%	3%	94%	0%	0%
Mubende	399.064	3.00	64.76	328.30	3.00	0.00	1%	16%	82%	1%	0%
Soroti	820.935	5.92	41.65	694.94	78.42	0.00	1%	5%	85%	10%	0%
Tororo	459.787	6.00	5.00	357.84	88.55	2.40	1%	1%	78%	19%	1%

Appendix 7: Road Condition distribution map, 2022

Appendix 8: Traffic distribution map, 2022

Appendix 9: Structures in critical condition based on overall rating

List of critical bridges on the network based on overall rating

S/N	Bridge No.	Bridge Name	Link ID	Link Name	Chainage (km)	Maintenance Station	Maintenance Region	Number of spans	Bridge Length	Bridge Width	Overall Rating
1	B016	Rwimi	A005_Link11	Kampala-Mubende-Fort	0	Fort Portal	Western	1	16	7	Poor
2	B080	Mubuku	C510_Link01	Mubuku-Maliba-Nyakalingijo	10.7	Kasese	Southern	1	26	5	Poor
3	B098	Nkola	C833_Link01	Nabumali junction-Busolwe	34.488	Tororo	Eastern	1	7	4	Poor
4	B104	Sonoli	C862_Link01	Namagumba - Budadiri	18.427	Mbale	Eastern	1	7	8	Poor
5	B105	Namatsu	C863_Link01	Mbale - Nkokonjeru	21.067	Mbale	Eastern	1	8	10	Poor
6	B109	Tsutsu	C867_Link02	Bumbobi-Bubulo-Budud	28.48	Mbale	Eastern	1	9	8	Poor
7	B150	Agora	B252_Link02	Lira-Acholibur-Kitgum	104.78	Kitgum	Northern	1	5	8	Poor
8	B176	Lakapelot	B301_Link03	Kotido - Apaan - Kabong	102.411	Kotido	North Eastern	1	16	7	Poor
9	B189	Awere	C680_Link02	Gulu-Opit-Rwackoko	74.91	Gulu	Northern	3	50	4	Poor
10	B215	Lojoro	C960_Link01	Angatun-Nabilatuk-Lokapel	46.817	Moroto	North Eastern	1	6	4	Poor
11	B323	Agagura	C682_Link01	Gulu - Logere - Adee		Gulu	Northern	2	9	5	Poor
12	B338	Acodho	C747_Link01	Panyimur -- Vurra Customs	18.264	Arua	Northern	1	7	4	Poor
13	B348	Kochi	C758_Link01	Keriri - Midigo	6.7	Moyo	Northern	4	24	4	Poor
14	B352	Korubude	C836_Link02	Mulanda - Nagongera		Tororo	Eastern	1	4	4	Poor
15	B354	Kaminima	C837_Link01	Nagongera - Merikit	18.218	Tororo	Eastern	1	5	5	Poor
16	B355	Kamunima	C837_Link01	Nagongera - Merikit	18.218	Tororo	Eastern	1	5	5	Poor
17	B362	Nahamya	C873_Link01	Nabiganda - Kaiti	3.436	Mbale	Eastern	2	15	4	Poor
18	B383	Omongo	C964_Link02	Toroma-Magoro	17.5	Soroti	North Eastern	2	12	4	Poor
19	B864	Kiguli	C862_Link01	Namagumba - Budadiri	23	Mbale	Eastern	1	10	7	Poor
20	B865	Leresi	C833_Link01	Nabumali junction-Busolwe	34.6	Tororo	Eastern	1	8	5	Poor
21	B874	Bukwo	B304_Link02	Kaserem-Kapchorwa-Suam	98.69	Mbale	Eastern	1	21	6	Poor
22	B875	Nangipa	C837_Link02	Kidoko - Busiu - Isikhoye	18.218	Tororo	Eastern	1	5	4	Poor
23	B876	Lukha	C837_Link02	Kidoko - Busiu - Isikhoye	18.218	Tororo	Eastern	1	6	5	Poor
24	B879	Lonene	C968_Link01	Matany - Lokopo - Turtuko	29.827	Moroto	North Eastern	2	19	3	Poor
25	B886	Kwakiteza	B306_Link01	Tororo (Malaba junction) - Busia	23	Tororo	Eastern	1	11	5	Poor

List of critical major culverts on the network based on overall rating

Culvert No	Link No	Link Name	Chainage (km)	Maintenance Station	Maintenance Region	Type of culvert	No. of pipes/ cells	Overall Rating	Span/ Diameter (m)
C011	C317_Link01	Katuugo-Kinyogogga	12	Luwero	Central	Corrugated metal pipe culvert	6	Poor	2.8
C558	C726_Link01	Adjumani - Sinyanya Ferry landing	14	Moyo	Northern	Concrete pipe culvert	3	Poor	1.8